AP CALCULUS BC · EXAMINATION REPOSITORY

Unit 8
Applications of Integration

100-Question Priority-Starred Master Set

- Questions 1–90: Multiple Choice (Full Priority Ratings · 4 Questions Per Page)
- Questions 91–100: Free Response with Official AP Scoring Rubrics (9 Pts Each)
- Integrated AP Vector Curve Diagrams, Cross Sections, Washers & Numerical Tables
- Complete 90-Question Answer Key & Step-by-Step Bulleted Solutions with Distractor Analysis

OFFICIAL AP CED ALIGNED · MEDIUM LEVEL · RIGOROUSLY VERIFIED

Examination Structure

This comprehensive paper contains 100 continuously numbered questions strictly covering Unit 8 (Applications of Integration), incorporating official AP-style vector graphs, tables of values, and geometric cross-section/revolution diagrams to test conceptual and analytical calculus mastery.

⭐ Priority Rating System

★★★★★ VERY IMPORTANT	High-frequency core exam topics: Average Value & MVT for Integrals, Particle Motion (Displacement vs. Distance), Area Between Curves, Volumes by Washer/Disc with Shifted Axes, and BC Arc Length.
★★★★ MANDATORY	Essential analytical skills: Volumes with Known Cross Sections (Squares, Semicircles, Triangles), Net Change Applied Accumulation Modeling, Slicing with Respect to y, and Planar Parametric Distance.
★★★ IMPORTANT	Foundational practice: Speed increasing/decreasing criteria, constant height rectangular cross sections, integral equation setups, and single-variable rate accumulations.

- **Questions 1–90:** Multiple Choice, four choices, exactly one correct answer (Exactly 4 MCQs per page).
- **Questions 91–100:** Free Response Questions, 9 points each, with authentic AP multi-part rubrics.
- **Visual Elements:** Crisp vector SVG area graphs, solids with known cross sections, washer revolution schematics, and numerical step tables.
- **Calculator Policy:** No calculator required (Closed-form analytical evaluation).
- **Answer Section:** Complete Answer Key grid, followed by step-by-step bulleted solutions and distractor breakdowns.

Topic Coverage (Unit 8.1 – 8.12)

8.1 Finding Average Value of a Function on an Interval	8.2 Connecting Position, Velocity & Acceleration via Integrals
8.3 Accumulation Functions & Applied Rate Contexts	8.4 Finding Area Between Curves: Functions of x
8.5 Finding Area Between Curves: Functions of y	8.6 Area Between Curves with Multiple Intersections
8.7 Volumes with Known Cross Sections (Squares, Triangles)	8.8 Volumes of Revolution: Disc Method
8.9 Volumes of Revolution: Washer Method	8.10 Volumes of Revolution: Shifted Axes (y = k, x = k)
8.11 BC: Arc Length of a Smooth Curve	8.12 BC: Distance Traveled by a Particle in a Plane

Essential Tip: When setting up washer volume integrals, always draw the axis of revolution first. Determine R(x) (distance from axis to the FARTHER curve) and r(x) (distance to the CLOSER curve), remembering that Volume = $\pi \int [R^2 - r^2]\, dx$, never $\pi \int (R - r)^2\, dx$!

ITEM 01

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the average value of the function $f(x) = 3x^2 - 2x + 1$ on the closed interval $[0, 3]$?

A 7 **B** 21

C 9 **D** 14

ITEM 02

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The table below gives values of a continuous function f at selected values of x.

x	1	3	5	7	9
f(x)	4	8	12	16	10

Using a Trapezoidal sum with the four subintervals indicated by the table, what is the approximation for the average value of f on the interval $[1, 9]$?

A 86.00 **B** 10.75

C 12.25 **D** 21.50

ITEM 03

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the average value of $f(x) = \sin x$ on the closed interval $[0, \pi]$?

A 0 **B** $1/\pi$

C $2/\pi$ **D** 2

ITEM 04

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Let $f(x) = x^2$. According to the Mean Value Theorem for Integrals, there exists a number c in the open interval $(0, 3)$ such that $f(c)$ equals the average value of f on $[0, 3]$. What is the exact value of c?

A 1.5 **B** $\sqrt{6}$

C 3 **D** $\sqrt{3}$

ITEM 05

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The temperature T (in °F) of an outdoor testing chamber over a 24-hour cycle is modeled by $T(t) = 50 + 10\sin(\pi t / 12)$, where t is measured in hours for $0 \le t \le 24$. What is the average temperature in the chamber over this 24-hour period?

A 50°F
B 55°F
C 60°F
D $50 + 10/\pi$ °F

ITEM 06

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the average value of $f(x) = \sqrt{x}$ on the interval $[0, 4]$?

A 8/3
B 4/3
C 16/3
D 2

ITEM 07

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

If the average value of a continuous function f on the interval $[1, 5]$ is equal to 6, what is the value of $\int_1^5 (2f(x) + 3)\,dx$?

A 27
B 36
C 60
D 48

ITEM 08

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the average value of $f(x) = e^{2x}$ on the closed interval $[0, \ln 2]$?

A $4 / \ln 2$
B $3 / \ln 2$
C $3 / 2$
D $3 / (2 \ln 2)$

ITEM 09

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A municipal water pump operates such that the rate of water consumption R(t) (in gallons per hour) is recorded at regular intervals:

t (hours)	0	2	4	6
R(t)	10	14	22	18

Using a Left Riemann sum with 3 equal subintervals, what is the approximation for the average rate of water consumption over the 6-hour period?

A 15.33 gal/hr

B 46.00 gal/hr

C 92.00 gal/hr

D 18.00 gal/hr

ITEM 10

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the average value of the function $f(x) = \sqrt{(16 - x^2)}$ on the interval $[-4, 4]$?

A 2π

B π

C 4π

D 8π

ITEM 11

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A particle moves along the x-axis with velocity $v(t) = 3t^2 - 12$ for $t \geq 0$. If the position of the particle at $t = 1$ is $x(1) = 5$, what is the position of the particle at $t = 3$?

A 14

B 9

C 7

D 3

ITEM 12

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A particle moves along a straight line with velocity $v(t) = 6 - 2t$ for $0 \leq t \leq 5$. What is the total distance traveled by the particle over the time interval $[0, 5]$?

A 5

B 9

C 15

D 13

ITEM 13

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The velocity v(t) of a particle moving along the y-axis is given by the table below for selected values of t in seconds.

t (sec)	0	2	5	8	10
v(t) (m/s)	3	7	4	−2	−6

Using a trapezoidal sum with the four subintervals indicated by the table, what is the approximation for the particle's displacement from t = 0 to t = 10?

A 21.5 m
B 16.5 m
C 31.5 m
D 12.0 m

ITEM 14

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

The acceleration of a particle moving along a line is given by $a(t) = 4e^{2t}$ for $t \geq 0$. If the particle's initial velocity is $v(0) = 3$, what is its velocity at $t = 2$?

A $2e^4 + 3$
B $2e^4 + 1$
C $4e^4 - 1$
D $8e^4 + 3$

ITEM 15

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A particle moves along the x-axis with velocity $v(t) = \cos(\pi t / 2)$ for $0 \leq t \leq 3$. If $x(0) = 0$, what is the total distance traveled by the particle over the interval $[0, 3]$?

A $2 / \pi$
B $4 / \pi$
C $6 / \pi$
D 0

ITEM 16

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

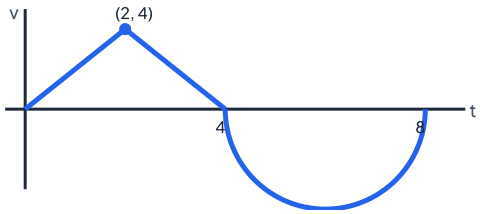
A particle moves along a coordinate axis such that its position is given by s(t) with $s(2) = -3$. Its velocity is $v(t) = t / \sqrt{(t^2 + 5)}$. What is the position of the particle at $t = 6$?

A $\sqrt{41} + 3$
B $\sqrt{41} - 3$
C $\sqrt{41}$
D $\sqrt{41} - 6$

ITEM 17

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The graph of the velocity v(t) of a particle moving along the x-axis for 0 ≤ t ≤ 8 consists of two line segments and a semicircle, as shown below.



If x(0) = 2, what is the position of the particle at t = 8?

A 10 – 2π
B 10 – 4π
C 8 – 2π
D 10 + 2π

ITEM 18

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

A particle travels along a straight path with velocity v(t) = 4t – 8 for t ≥ 0. During which time interval is the speed of the particle decreasing?

A t > 2
B 0 < t < 2
C t > 0
D Never

ITEM 19

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A particle moves along the x-axis with acceleration a(t) = 12t – 4. At t = 1, its velocity is v(1) = 3 and its position is x(1) = 2. What is the particle's position at t = 2?

A 13
B 9
C 11
D 15

ITEM 20

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The displacement of a particle moving along a straight line over the time interval [0, 4] is 8 meters. If the particle's velocity is given by v(t) = k · t for some constant k, and the total distance traveled by the particle over [0, 4] is also 8 meters, what is the value of k?

A 4
B 2
C 0.5
D 1

ITEM 21

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Water is pumped into a tank at a rate modeled by $R(t) = 60 - 4t$ liters per minute and leaves the tank at a constant rate of 20 liters per minute for $0 \le t \le 8$. At $t = 0$, the tank contains 150 liters of water. How much water is in the tank at $t = 5$?

A 300 liters
B 250 liters
C 350 liters
D 200 liters

ITEM 22

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

Oil leaks from a storage tank at a rate of R(t) liters per hour, where t is measured in hours. Values of R(t) are recorded in the table below at selected times.

t (hours)	0	2	5	9	12
R(t) (L/hr)	18	15	12	8	5

Using a Left Riemann sum with the four subintervals indicated by the table, what is the approximation for the total volume of oil that leaked from the tank between $t = 0$ and $t = 12$ hours?

A 134 liters
B 153 liters
C 113 liters
D 166 liters

ITEM 23

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Customers enter an amusement park at a rate of $E(t) = 120t - 15t^2$ customers per hour and exit at a rate of $L(t) = 40t$ customers per hour for $0 \le t \le 8$ hours after opening ($t = 0$). Assuming there are no customers in the park at $t = 0$, at what time t is the number of customers inside the park at its maximum?

A $t = 4$ hours
B $t = 8/3$ hours
C $t = 16/3$ hours
D $t = 8$ hours

ITEM 24

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A pollutant is removed from a lake at a rate of $R(t) = 80 / (t + 4)^2$ kg per day for $t \ge 0$. What is the total amount of pollutant removed from the lake over the first 6 days (from $t = 0$ to $t = 6$)?

A 16 kg
B 20 kg
C 8 kg
D 12 kg

ITEM 25

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

The rate of electricity consumption in a manufacturing plant is given by $C(t) = 10 + 6\sin(\pi t / 12)$ megawatts, where t is measured in hours since midnight. What is the total electricity consumed by the plant over the first 12 hours of the day?

A $120 + 144/\pi$ MW·hr

B $120 + 72/\pi$ MW·hr

C 120 MW·hr

D $120 - 144/\pi$ MW·hr

ITEM 26

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A bacteria colony grows at a rate of $R(t) = 300 \cdot e^{0.2t}$ bacteria per hour. If the initial population at $t = 0$ is 1,200 bacteria, what is the colony population at $t = 5$ hours?

A $1{,}500\,e$

B $1{,}500\,e - 300$

C $1{,}500\,e - 1{,}500$

D $1{,}200\,e$

ITEM 27

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A heater warms a room at a rate H(t) °C/hr while heat escapes through the windows at a rate W(t) °C/hr. If the room temperature is 18°C at $t = 0$, and the net rate of temperature change $H(t) - W(t) = 4 - 0.5t$ for $0 \le t \le 6$, what is the room temperature at $t = 6$ hours?

A 27°C

B 15°C

C 33°C

D 30°C

ITEM 28

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

Gravel is added to a pile at a rate of $G(t) = 12 + 6t^2$ tons per hour. How many tons of gravel are added to the pile during the time interval from $t = 1$ to $t = 3$ hours?

A 52 tons

B 64 tons

C 80 tons

D 76 tons

ITEM 29

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

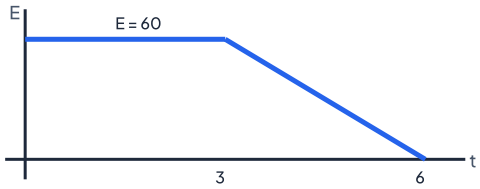
A metal rod of length 4 meters has linear mass density $\rho(x) = 3 + 2x - 0.5x^2$ kg/meter, where x is the distance in meters from the left end. What is the total mass of the rod?

A 17.33 kg
B 16.00 kg
C 20.67 kg
D 18.50 kg

ITEM 30

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The rate at which people enter an auditorium is modeled by the function E(t) shown in the graph below, consisting of a horizontal line and a line segment for $0 \le t \le 6$ hours.



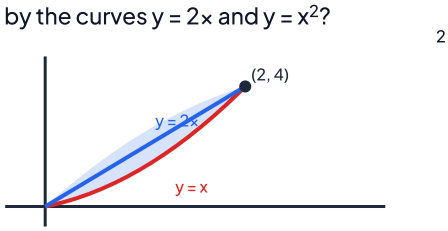
How many people entered the auditorium during the 6-hour period?

A 360 people
B 270 people
C 180 people
D 240 people

ITEM 31

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the area of the region bounded by the curves $y = 2x$ and $y = x^2$?



A 2
B 2 / 3
C 4 / 3
D 8 / 3

ITEM 32

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the area of the region enclosed between the curves $y = \cos x$ and $y = \sin x$ from $x = 0$ to $x = \pi / 4$?

A $2 - \sqrt{2}$
B $\sqrt{2}$
C $1 - \sqrt{2} / 2$
D $\sqrt{2} - 1$

ITEM 33

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the area of the region bounded by the parabola $y = 4 - x^2$ and the line $y = x + 2$?

A 9 / 2
B 7 / 2
C 16 / 3
D 9

ITEM 34

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the area of the region bounded by $y = e^x$ and $y = e^{-x}$ over the interval $[0, \ln 3]$?

A 8 / 3
B 4 / 3
C 2
D 10 / 3

ITEM 35

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the area of the region bounded by $y = \sqrt{x}$ and $y = x / 2$?

A 8 / 3
B 2 / 3
C 4 / 3
D 2

ITEM 36

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the area of the region in the first quadrant bounded by $y = x^3 - 3x$ and the line $y = x$?

A 6
B 2
C 8
D 4

ITEM 37

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The vertical line x = k divides the region bounded by $y = x^2$ and the x-axis from x = 0 to x = 2 into two subregions of equal area. What is the value of k?

A 4^(1/3)
B 2^(1/3)
C 2 / √3
D 1

ITEM 38

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

What is the area of the region bounded by y = 1 / x, the x-axis, and the vertical lines x = 1 and $x = e^2$?

A 1
B 2
C $e^2 - 1$
D 2e

ITEM 39

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the area of the region enclosed by the parabola $y = 3 - x^2$ and the horizontal line y = −1?

A 8 / 3
B 16 / 3
C 32 / 3
D 64 / 3

ITEM 40

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The functions f and g are continuous for all x, with f(x) ≥ g(x) on [0, 6]. Selected values are given below.

x	0	2	4	6
f(x)	10	14	11	8
g(x)	2	4	5	4

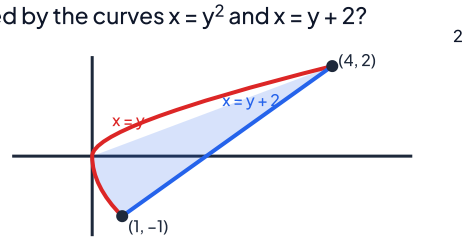
Using a trapezoidal sum with the three subintervals shown, what is the approximation for the area between the graphs of f and g on [0, 6]?

A 22
B 38
C 50
D 44

ITEM 41

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the area of the region bounded by the curves $x = y^2$ and $x = y + 2$?



A 9 / 2 **B** 7 / 2

C 16 / 3 **D** 9

ITEM 42

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the area of the region bounded by $x = 4 - y^2$ and the y-axis ($x = 0$)?

A 16 / 3 **B** 32 / 3

C 8 / 3 **D** 64 / 3

ITEM 43

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the area of the region bounded by the curves $x = e^y$, the line $x = 1$, and the horizontal line $y = 2$?

A $e^2 - 2$ **B** $e^2 - 1$

C $e^2 - 3$ **D** $2e^2 - 2$

ITEM 44

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the total area of the region enclosed between the curve $y = x^3 - x$ and the x-axis on the interval $[-1, 1]$?

A 1 **B** 0

C 1 / 4 **D** 1 / 2

ITEM 45

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the total area of the region enclosed between the curves $y = \sin x$ and $y = \cos x$ on the interval $[0, \pi]$?

A $2\sqrt{2}$
B $2\sqrt{2} - 2$
C $\sqrt{2}$
D 2

ITEM 46

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the area of the region bounded by the parabola $x = 2y - y^2$ and the y-axis ($x = 0$)?

A 2 / 3
B 4 / 3
C 8 / 3
D 2

ITEM 47

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The horizontal line $y = c$ divides the region bounded by $x = y^3$, the y-axis, and the horizontal line $y = 2$ into two subregions of equal area. What is the value of c?

A 1
B 2^(1/4)
C 2^(3/4)
D √2

ITEM 48

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the area of the region bounded by the curves $x = \sqrt{y}$ and $x = y^2$?

A 1 / 6
B 2 / 3
C 1 / 2
D 1 / 3

ITEM 49

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the total area of the region enclosed between the curves $y = x^3 - 2x^2$ and $y = 2x - x^2$?

A 37 / 12
B 35 / 12
C 8 / 3
D 5 / 12

ITEM 50

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The curves $x = f(y)$ and $x = g(y)$ are continuous for $1 \le y \le 5$, with $f(y) \ge g(y)$. Selected values are given below.

y	1	2	3	4	5
f(y)	7	9	11	8	6
g(y)	1	3	4	2	1

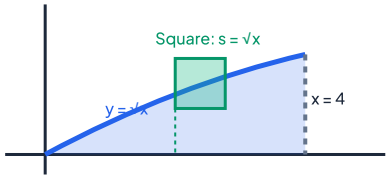
Using a trapezoidal sum with the four subintervals shown, what is the approximation for the area of the region bounded by $x = f(y)$ and $x = g(y)$ from $y = 1$ to $y = 5$?

A 27.0
B 24.5
C 31.5
D 18.0

ITEM 51

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The base of a solid is the region bounded by $y = \sqrt{x}$, the x-axis, and the line $x = 4$. Cross sections perpendicular to the x-axis are squares. What is the volume of the solid?



A 32 / 3
B 16
C 8
D 4

ITEM 52

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The base of a solid is the region enclosed by the parabola $y = 1 - x^2$ and the x-axis. Cross sections of the solid perpendicular to the x-axis are equilateral triangles. What is the volume of the solid?

A √3 / 3
B 8√3 / 15
C 2√3 / 5
D 4√3 / 15

ITEM 53

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The base of a solid is the region bounded by $y = \sin x$ and the x-axis on the interval $[0, \pi]$. Cross sections of the solid perpendicular to the x-axis are semicircles whose diameters lie in the base. What is the volume of the solid?

A $\pi^2 / 16$

B $\pi^2 / 8$

C $\pi^2 / 4$

D $\pi / 8$

ITEM 54

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The base of a solid is bounded by $y = e^x$, the x-axis, the y-axis, and the line $x = 1$. Cross sections perpendicular to the x-axis are squares. What is the volume of the solid?

A $e^2 - 1$

B $(e^2 - 1) / 2$

C $(e^2 - 1) / 4$

D $e^2 / 2$

ITEM 55

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The base of a solid is the region bounded by the parabola $y = x^2$ and the line $y = 4$. Cross sections perpendicular to the y-axis are squares. What is the volume of the solid?

A 16

B 64

C 32

D $128 / 3$

ITEM 56

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The base of a solid is the triangle bounded by the line $y = 2 - x$, the x-axis, and the y-axis. Cross sections perpendicular to the x-axis are isosceles right triangles with hypotenuse in the base. What is the volume of the solid?

A $8 / 3$

B $4 / 3$

C $1 / 3$

D $2 / 3$

ITEM 57

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

The base of a solid is bounded by $y = 2 \sin x$ and the x-axis on $[0, \pi]$. Cross sections perpendicular to the x-axis are rectangles of constant height 3. What is the volume of the solid?

A 12 **B** 6

C 18 **D** 6π

ITEM 58

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The base of a solid is the circular disk $x^2 + y^2 \le 9$. Cross sections of the solid perpendicular to the x-axis are squares. What is the volume of the solid?

A 72 **B** 144

C 36π **D** 288

ITEM 59

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A solid has a base along the x-axis for $0 \le x \le 8$. Cross sections perpendicular to the x-axis have areas A(x) given in the table below.

x	0	2	4	6	8
A(x)	3	7	12	10	4

Using a trapezoidal sum with the four subintervals indicated by the table, what is the approximation for the volume of the solid?

A 72 **B** 130

C 65 **D** 58

ITEM 60

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The base of a solid is bounded by $y = \ln x$, the x-axis, and the vertical line $x = e$. Cross sections perpendicular to the x-axis are isosceles right triangles with one leg in the base. What is the volume of the solid?

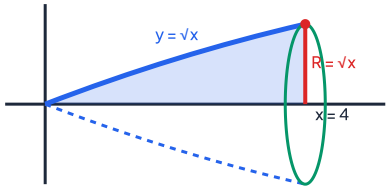
A $e / 2$ **B** $e - 2$

C $(e - 1) / 2$ **D** $(e - 2) / 2$

ITEM 61

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the volume of the solid generated when the region bounded by $y = \sqrt{x}$, the x-axis, and the line $x = 4$ is revolved about the x-axis?



A 8π
B 16π
C 4π
D $32\pi / 3$

ITEM 62

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the volume of the solid generated when the region bounded by $y = x^2$, the y-axis, and the horizontal line $y = 4$ is revolved about the y-axis?

A 16π
B 8π
C $32\pi / 3$
D $64\pi / 5$

ITEM 63

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the volume of the solid formed by revolving the region bounded by $y = e^x$, the x-axis, the y-axis, and the vertical line $x = 1$ about the x-axis?

A $\pi e^2 / 2$
B $\pi(e^2 - 1)$
C $\pi(e^2 - 1) / 2$
D $2\pi(e^2 - 1)$

ITEM 64

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the volume of the cone formed by revolving the triangle bounded by the line $y = 2 - x$, the x-axis, and the y-axis about the x-axis?

A $16\pi / 3$
B $4\pi / 3$
C 8π
D $8\pi / 3$

ITEM 65

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the volume of the solid generated by revolving the region bounded by $y = \cos x$ and the x-axis from $x = 0$ to $x = \pi / 2$ about the x-axis?

A $\pi^2 / 4$
B $\pi / 2$
C $\pi^2 / 2$
D $\pi / 4$

ITEM 66

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the volume of the sphere generated by revolving the semicircular region bounded by $y = \sqrt{(4 - x^2)}$ and the x-axis about the x-axis?

A $16\pi / 3$
B $32\pi / 3$
C 16π
D $64\pi / 3$

ITEM 67

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the volume of the solid generated when the region bounded by $y = 4 - x^2$ and the x-axis is revolved about the horizontal line $y = 0$?

A $128\pi / 5$
B $256\pi / 15$
C $512\pi / 15$
D $64\pi / 3$

ITEM 68

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the volume of the solid generated by revolving the region bounded by $x = 1 / y$, the y-axis, and the horizontal lines $y = 1$ and $y = 3$ about the y-axis?

A $\pi \ln 3$
B $\pi / 3$
C $4\pi / 3$
D $2\pi / 3$

ITEM 69

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

What is the volume of the cone formed by revolving the region bounded by $y = 3 - x$, the x-axis, and the y-axis about the y-axis?

A 9π B 27π

C 18π D $9\pi / 2$

ITEM 70

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A solid of revolution is generated by revolving a region about the x-axis for $0 \le x \le 6$. The radius R(x) of the cross-sectional disk at selected values of x is given in the table below.

x	0	2	4	6
R(x)	1	3	4	2

Using a trapezoidal sum with the three subintervals shown, what is the approximation for the volume of the solid?

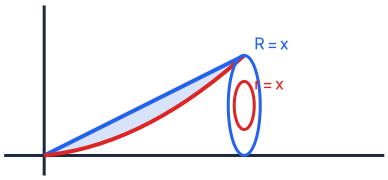
A 110π B 55π

C 60π D 50π

ITEM 71

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the volume of the solid generated when the region bounded by $y = x$ and $y = x^2$ is revolved about the x-axis?



A $4\pi / 15$ B $\pi / 15$

C $2\pi / 15$ D $\pi / 3$

ITEM 72

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the volume of the solid generated when the region bounded by $y = x$ and $y = x^2$ is revolved about the y-axis?

A $\pi / 12$ B $\pi / 3$

C $2\pi / 15$ D $\pi / 6$

ITEM 73

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The region bounded by $y = 2x$ and $y = x^2$ is revolved about the horizontal line $y = 5$. What is the volume of the resulting solid?

A $136\pi / 15$
B $68\pi / 15$
C $32\pi / 5$
D $128\pi / 15$

ITEM 74

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The region bounded by $y = 4 - x^2$ and the x-axis is revolved about the horizontal line $y = -1$. What is the volume of the resulting solid?

A $512\pi / 15$
B $832\pi / 15$
C $416\pi / 15$
D $256\pi / 5$

ITEM 75

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The region in the first quadrant bounded by $x = y^2$ and $x = 2y$ is revolved about the vertical line $x = 5$. What is the volume of the resulting solid?

A $112\pi / 15$
B $68\pi / 15$
C $136\pi / 15$
D $128\pi / 15$

ITEM 76

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

What is the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$ and $y = x / 2$ about the x-axis?

A 8π
B $4\pi / 3$
C $16\pi / 3$
D $8\pi / 3$

ITEM 77

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

The region bounded by $y = e^x$, the horizontal line $y = 1$, and the vertical line $x = 1$ is revolved about the horizontal line $y = -2$. Which expression gives the exact volume of the resulting solid?

A $\pi(e^2 + 8e - 19) / 2$

B $\pi(e^2 + 4e - 9) / 2$

C $\pi(e^2 - 1) / 2$

D $\pi(e^2 + 8e - 9)$

ITEM 78

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

The region bounded by $y = 1 / x$, the x-axis, and the vertical lines $x = 1$ and $x = 2$ is revolved about the horizontal line $y = -1$. What is the volume of the solid generated?

A $\pi(1 / 2 + \ln 2)$

B $\pi(1 / 2 + 2 \ln 2)$

C $\pi(3 / 4 + 2 \ln 2)$

D $\pi(1 + 2 \ln 2)$

ITEM 79

EASY · SINGLE BEST CHOICE ★★★ IMPORTANT

Which of the following definite integrals represents the volume of the solid generated when the region bounded by $y = \sin x$ and the x-axis on $[0, \pi]$ is revolved about the line $y = -2$?

A $\pi \int_0^{\pi} [(\sin x)^2 - 4]\, dx$

B $\pi \int_0^{\pi} (\sin x + 2)^2\, dx$

C $\pi \int_0^{\pi} [(\sin x + 2)^2 - 4]\, dx$

D $\pi \int_0^{\pi} (\sin x - 2)^2\, dx$

ITEM 80

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A solid of revolution has outer radius R(x) and inner radius r(x) with values given in the table below.

x	1	2	3	4	5
R(x)	5	6	7	6	5
r(x)	2	2	3	2	1

Using a trapezoidal sum with the four subintervals shown, what is the approximation for the volume of the solid generated when revolved about the x-axis?

A 138.5π

B 253.0π

C 115.0π

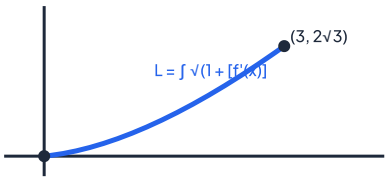
D 126.5π

ITEM 81

EASY · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the length of the curve y = (2/3) x^(3/2) from x = 0 to x = 3?

^{2}) dxx = 0x = 3



A 14 / 3

B 16 / 3

C 7 / 3

D 8

ITEM 82

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the length of the curve y = (x^3 / 6) + (1 / (2x)) on the closed interval [1, 2]?

A 13 / 12

B 17 / 12

C 19 / 12

D 3 / 2

ITEM 83

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

Which of the following definite integrals represents the total length of the curve y = sin(2x) from x = 0 to x = π?

A $\int_0^{\pi} \sqrt{(1 + 2\cos(2x))}\,dx$

B $\int_0^{\pi} \sqrt{(1 + \cos^2(2x))}\,dx$

C $\int_0^{\pi} \sqrt{(1 + 4\cos^2(2x))}\,dx$

D $\int_0^{\pi} \sqrt{(1 + 4\sin^2(2x))}\,dx$

ITEM 84

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the length of the curve x = (1/3) (y^2 + 2)^(3/2) from y = 0 to y = 1?

A 1

B 2 / 3

C 5 / 3

D 4 / 3

ITEM 85

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

A particle moves in the xy-plane with parametric equations $x(t) = 3\cos t$ and $y(t) = 3\sin t$ for $0 \le t \le 2\pi$. What is the total distance traveled by the particle?

A 6π **B** 3π

C 9π **D** 12π

ITEM 86

HARD · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

A particle moves in the xy-plane with velocity vector $v(t) = \langle 2t, 3t^2 \rangle$ for $0 \le t \le 2$. What is the total distance traveled by the particle over the time interval $[0, 2]$?

A $(4 / 27)(10\sqrt{10} - 1)$ **B** $(8 / 27)(10\sqrt{10} - 1)$

C $(8 / 9)(10\sqrt{10} - 1)$ **D** $(16 / 27)(10\sqrt{10} - 1)$

ITEM 87

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

What is the length of the curve $y = \ln(\cos x)$ from $x = 0$ to $x = \pi / 3$?

A $\ln 2$ **B** $\ln(1 + \sqrt{3})$

C $\ln(2 + \sqrt{3})$ **D** $2\ln(2 + \sqrt{3})$

ITEM 88

EASY · SINGLE BEST CHOICE ★★★★ MANDATORY

A particle moves in the xy-plane with velocity components $x'(t) = e^t \cos t$ and $y'(t) = e^t \sin t$ for $0 \le t \le \ln 4$. What is the total distance traveled by the particle?

A 2 **B** 4

C $\ln 4$ **D** 3

ITEM 89

MODERATE · SINGLE BEST CHOICE ★★★★ MANDATORY

A smooth curve $y = f(x)$ has derivative values $f'(x)$ given in the table below.

x	0	1	2	3	4
$f'(x)$	0	$\sqrt{3}$	$\sqrt{8}$	$\sqrt{15}$	$\sqrt{24}$

Using a trapezoidal sum with the four subintervals indicated by the table, what is the approximation for the arc length of the curve from $x = 0$ to $x = 4$?

A 12

B 24

C 10

D 14

ITEM 90

MODERATE · SINGLE BEST CHOICE ★★★★★ VERY IMPORTANT

Consider the region R bounded by $y = x^2$, the x-axis, and $x = 1$. Which of the following statements correctly orders the area of R, the volume V_1 formed by revolving R about the x-axis, and the arc length L of $y = x^2$ from $x = 0$ to $x = 1$?

A $V_1 < \text{Area} < L$

B $\text{Area} < V_1 < L$

C $\text{Area} < L < V_1$

D $L < \text{Area} < V_1$

FRQ 1: Particle Motion on a Coordinate Line (Displacement, Distance, Tables & Turning Points)

★★★★★ VERY IMPORTANT | 9 Points

A particle moves along a straight horizontal axis with velocity v(t), measured in meters per second, for $0 \le t \le 10$ seconds. Selected values of v(t) are recorded in the table below.

t (sec)	0	2	5	7	10
v(t) (m/s)	5	8	2	−4	−1

The particle's position at time $t = 0$ is $x(0) = -3$ meters.

(A) [2 Points] Use the data in the table to estimate the acceleration of the particle at time $t = 6$ seconds. Show the computations that lead to your answer and indicate units of measure.

(B) [2 Points] Using a trapezoidal sum with the four subintervals indicated by the table, approximate the displacement of the particle over the time interval $0 \le t \le 10$ seconds.

(C) [3 Points] For $0 \le t \le 2$, the particle's velocity is modeled by $v_1(t) = 5 + 3t - (3/4)t^2$. Find the exact position of the particle at $t = 2$.

(D) [2 Points] Explain why there must be at least one time t in the interval $(2, 7)$ at which the particle is at rest ($v(t) = 0$). Must the particle's total distance traveled on $[0, 10]$ be strictly greater than its displacement? Justify your reasoning.

Official AP Scoring Directive: Show all supporting work. Clearly identify limits, integrands, and antiderivatives. Express answers in simplest exact analytical form.

FRQ 2: Water Storage Inflow and Outflow Rates (Net Change & Optimization)

★★★★★ VERY IMPORTANT

9 Points

Water is pumped into a municipal storage reservoir at a rate modeled by $R(t) = 60 - 6t$ gallons per minute, for $0 \le t \le 8$ minutes. Water is removed from the reservoir for distribution at a constant rate of $D(t) = 24$ gallons per minute for $0 \le t \le 8$. At time $t = 0$, the reservoir contains 200 gallons of water.

(A) [2 Points] Find the total amount of water pumped into the reservoir between time $t = 0$ and $t = 8$ minutes.

(B) [2 Points] Is the amount of water in the reservoir increasing or decreasing at time $t = 5$ minutes? Give a reason for your answer.

(C) [3 Points] Find the absolute maximum amount of water in the reservoir, in gallons, over the time interval $0 \le t \le 8$. Justify your answer.

(D) [2 Points] Write, but do not evaluate, an expression involving an integral that gives the average rate of inflow of water into the reservoir over the interval $[0, 8]$.

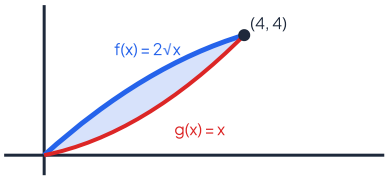
Official AP Scoring Directive: Show all supporting work. Clearly identify limits, integrands, and antiderivatives. Express answers in simplest exact analytical form.

FRQ 3: Planar Region Bounded by Curves, Area & Dividing Lines

★★★★★ VERY IMPORTANT

9 Points

Let R be the region in the first quadrant enclosed by the graphs of $f(x) = 2\sqrt{x}$ and $g(x) = (1/2)x^2$.



(A) [3 Points] Find the points of intersection of f(x) and g(x), and calculate the area of region R.

(B) [3 Points] The vertical line x = k divides region R into two subregions of equal area. Write, but do not solve, an equation involving one or more integrals that can be used to find the value of k.

(C) [3 Points] Express the area of region R by integrating with respect to y. Show that this yields the identical area calculated in Part (A).

Official AP Scoring Directive: Show all supporting work. Clearly identify limits, integrands, and antiderivatives. Express answers in simplest exact analytical form.

FRQ 4: Volumes of Solids with Known Cross Sections

★★★★★ VERY IMPORTANT 9 Points

Let R be the region enclosed by the parabola $y = 4 - x^2$ and the x-axis for $-2 \le x \le 2$.

(A) [3 Points] The region R is the base of a solid. Cross sections perpendicular to the x-axis are squares. Find the volume of this solid.

(B) [3 Points] The region R is the base of another solid. Cross sections perpendicular to the x-axis are equilateral triangles. Find the volume of this solid.

(C) [3 Points] The region R is the base of a third solid. Cross sections perpendicular to the x-axis are semicircles whose diameters lie in the base. Find the volume of this solid.

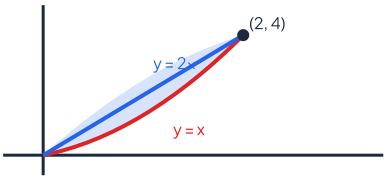
Official AP Scoring Directive: Show all supporting work. Clearly identify limits, integrands, and antiderivatives. Express answers in simplest exact analytical form.

FRQ 5: Volumes of Revolution: Washer Method around Horizontal & Vertical Axes

★★★★★ VERY IMPORTANT

9 Points

Let R be the region in the first quadrant bounded by the curves $y = 2x$ and $y = x^2$.



(A) [3 Points] Find the volume of the solid generated when R is revolved about the x-axis ($y = 0$).

(B) [3 Points] Find the volume of the solid generated when R is revolved about the horizontal line $y = 5$.

(C) [3 Points] Write, but do not evaluate, an integral expression that gives the volume of the solid generated when R is revolved about the vertical line $x = 3$.

Official AP Scoring Directive: Show all supporting work. Clearly identify limits, integrands, and antiderivatives. Express answers in simplest exact analytical form.

FRQ 6: BC Exclusive: Arc Length, Perimeter & Volume of a Planar Solid

★★★★★ VERY IMPORTANT

9 Points

Let R be the region bounded by the curve y = (2/3) x^(3/2), the horizontal line y = 0, and the vertical line x = 3.

(A) [2 Points] Find the area of region R.

(B) [4 Points] Find the exact perimeter of region R, including the curved portion and all straight boundaries.

(C) [3 Points] Find the volume of the solid generated when region R is revolved about the x-axis.

Official AP Scoring Directive: Show all supporting work. Clearly identify limits, integrands, and antiderivatives. Express answers in simplest exact analytical form.

FRQ 7: Comprehensive Solid Modeling - Cross Sections vs. Revolution

★★★★★ VERY IMPORTANT | 9 Points

Let R be the region bounded by the curves $y = e^x$, the line $y = 1$, and the vertical line $x = \ln 4$.

(A) [2 Points] Find the area of region R.

(B) [3 Points] The region R is the base of a solid. Cross sections perpendicular to the x-axis are rectangles whose height is twice the length of the base in the xy-plane. Find the volume of the solid.

(C) [4 Points] Find the volume of the solid generated when region R is revolved about the horizontal line $y = 1$.

Official AP Scoring Directive: Show all supporting work. Clearly identify limits, integrands, and antiderivatives. Express answers in simplest exact analytical form.

FRQ 8: BC Exclusive: Planar Motion, Parametric Arc Length & Distance Traveled

★★★★★ VERY IMPORTANT

9 Points

A particle moves in the xy-plane so that its position at any time $t \geq 0$ is given by $(x(t), y(t))$. The particle's velocity vector is $v(t) = \langle 4t, 3t^2 \rangle$. At time $t = 0$, the particle is at the point $(2, -5)$.

(A) [2 Points] Find the speed of the particle at time $t = 2$.

(B) [2 Points] Find the acceleration vector of the particle at time $t = 2$.

(C) [3 Points] Find the total distance traveled by the particle over the time interval $0 \leq t \leq 3$.

(D) [2 Points] Find the coordinates of the particle's position at time $t = 3$.

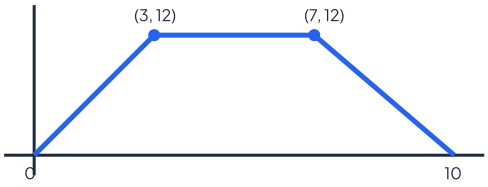
Official AP Scoring Directive: Show all supporting work. Clearly identify limits, integrands, and antiderivatives. Express answers in simplest exact analytical form.

FRQ 9: Applied Accumulation & Geometric Rate Function Analysis

★★★★★ VERY IMPORTANT

9 Points

The continuous function r(t), shown below for $0 \le t \le 10$, models the rate at which fuel is consumed by an engine in liters per hour. The graph consists of three line segments.



At t = 0, there are 150 liters of fuel in the tank.

(A) [2 Points] Find the total amount of fuel consumed by the engine over the time interval $0 \le t \le 10$ hours.

(B) [2 Points] Find the amount of fuel remaining in the tank at time t = 7 hours.

(C) [3 Points] Find the average rate of fuel consumption, in liters per hour, over the interval [0, 10].

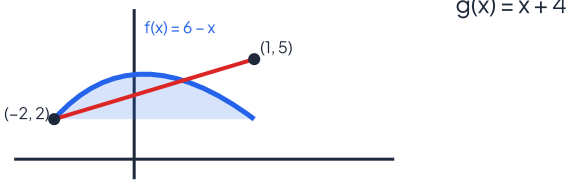
(D) [2 Points] At what time t in the interval (0, 10) does the instantaneous rate of fuel consumption equal the average rate of fuel consumption on [0, 10]? Justify your answer.

Official AP Scoring Directive: Show all supporting work. Clearly identify limits, integrands, and antiderivatives. Express answers in simplest exact analytical form.

FRQ 10: Full Unit 8 Capstone (Area, Washer Volume, Cross Sections & Arc Length)

★★★★★ VERY IMPORTANT | 9 Points

Let R be the region bounded by the curves $f(x) = 6 - x^2$ and $g(x) = x + 4$.



^{2}g(x) = x + 4

(A) [2 Points] Find the points of intersection of f(x) and g(x), and determine the area of region R.

(B) [3 Points] Find the volume of the solid generated when region R is revolved about the horizontal line y = 7.

(C) [2 Points] Region R is the base of a solid. Cross sections perpendicular to the x-axis are squares. Write, but do not evaluate, an integral expression for the volume of this solid.

(D) [2 Points] Write, but do not evaluate, an expression involving one or more integrals that gives the perimeter of region R.

Official AP Scoring Directive: Show all supporting work. Clearly identify limits, integrands, and antiderivatives. Express answers in simplest exact analytical form.

End of Question Paper

You have concluded Questions 1 through 100 of the Unit 8 Master Examination Repository.

> **Important Study Strategy:** Carefully compare your solutions against the detailed answer keys and explanations that follow. Pay special attention to why incorrect choices were selected as distractors and how full credit is awarded on official AP rubrics.

Answer Section Contents

- **Answer Key Table:** Complete quick-reference grid of letter choices for Questions 1–90.
- **MCQ Detailed Solutions:** Step-by-step bulleted solutions for Questions 1–90 with full distractor error analysis.
- **FRQ Model Solutions:** Full analytical solutions for Questions 91–100 with official AP 9-point rubric point breakdowns.
- **Formula Appendix:** Summary of high-yield Unit 8 integration applications, volume methods, and arc length formulas.

Multiple Choice Answer Key

Verified answer keys with priority ratings. Questions 91–100 are free response (90 points total).

Q01	Q02	Q03	Q04	Q05	Q06	Q07	Q08	Q09	Q10
A ★★★★★	B ★★★★★	C ★★★★	D ★★★★★	A ★★★★	B ★★★★	C ★★★★★	D ★★★★	A ★★★★★	B ★★★★★
Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
C ★★★★★	D ★★★★★	A ★★★★	B ★★★★	C ★★★★★	D ★★★★	A ★★★★★	B ★★★	C ★★★★★	D ★★★★
Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30
A ★★★★★	B ★★★★	C ★★★★★	D ★★★★	A ★★★	B ★★★★★	C ★★★★	D ★★★★	A ★★★★★	B ★★★★
Q31	Q32	Q33	Q34	Q35	Q36	Q37	Q38	Q39	Q40
C ★★★★★	D ★★★★	A ★★★★★	B ★★★★	C ★★★★	D ★★★★★	A ★★★★★	B ★★★	C ★★★★	D ★★★★★
Q41	Q42	Q43	Q44	Q45	Q46	Q47	Q48	Q49	Q50
A ★★★★★	B ★★★★	C ★★★★	D ★★★★★	A ★★★★★	B ★★★★	C ★★★★★	D ★★★★	A ★★★★★	B ★★★★
Q51	Q52	Q53	Q54	Q55	Q56	Q57	Q58	Q59	Q60
C ★★★★★	D ★★★★★	A ★★★★★	B ★★★★	C ★★★★★	D ★★★★	A ★★★	B ★★★★★	C ★★★★	D ★★★★★
Q61	Q62	Q63	Q64	Q65	Q66	Q67	Q68	Q69	Q70
A ★★★★★	B ★★★★★	C ★★★★	D ★★★★	A ★★★★★	B ★★★★	C ★★★★★	D ★★★★	A ★★★	B ★★★★
Q71	Q72	Q73	Q74	Q75	Q76	Q77	Q78	Q79	Q80
C ★★★★★	D ★★★★★	A ★★★★★	B ★★★★★	C ★★★★★	D ★★★★	A ★★★★★	B ★★★★	C ★★★	D ★★★★
Q81	Q82	Q83	Q84	Q85	Q86	Q87	Q88	Q89	Q90
A ★★★★★	B ★★★★★	C ★★★★	D ★★★★★	A ★★★★	B ★★★★★	C ★★★★★	D ★★★★	A ★★★★	B ★★★★★

Scoring Summary: Each multiple choice question is worth 1 point (90 points total). Each free response question is worth 9 points (90 points total). Total Exam Value: 180 Points.

Question 01 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 7

STEP-BY-STEP SOLUTION

- **Recall the Average Value Formula:** The average value of a continuous function f on [a, b] is given by $f_{avg} = (1/(b-a)) \int_a^b f(x)\,dx$.
- **Identify Limits of Integration:** Here, a = 0 and b = 3, so b – a = 3 – 0 = 3.
- **Evaluate the Definite Integral:**
- • $\int_0^3 (3x^2 - 2x + 1)\,dx = [x^3 - x^2 + x]_0^3$.
- • At upper limit x = 3: $3^3 - 3^2 + 3 = 27 - 9 + 3 = 21$.
- • At lower limit x = 0: $0^3 - 0^2 + 0 = 0$.
- • Value of integral = 21 – 0 = 21.
- **Compute Average Value:** $f_{avg} = (1/3) \cdot 21 = 7$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Evaluates the definite integral $\int_0^3 f(x)\,dx = 21$ but forgets to divide by the interval length (b – a) = 3.
✗ **(C)** Arithmetic error in evaluating the polynomial at x = 3.
✗ **(D)** Divides 21 by 1.5 or averages f(0) = 1 and f(3) = 22 to get 11.5, rounded incorrectly.

Question 02 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 10.75

STEP-BY-STEP SOLUTION

- **Understand the Objective:** Approximate $f_{avg} = (1/(9-1)) \int_1^9 f(x)\,dx = (1/8) \int_1^9 f(x)\,dx$.
- **Subinterval Widths:** The grid points are spaced with equal step size Δx = 2 (intervals [1,3], [3,5], [5,7], [7,9]).
- **Apply the Trapezoidal Rule:**
- • $\int_1^9 f(x)\,dx \approx (\Delta x/2) \cdot [f(1) + 2f(3) + 2f(5) + 2f(7) + f(9)]$.
- • $(2/2) \cdot [4 + 2(8) + 2(12) + 2(16) + 10] = 1 \cdot [4 + 16 + 24 + 32 + 10] = 86$.
- **Divide by the Interval Length:**
- • $f_{avg} \approx (1/8) \cdot 86 = 86/8 = 10.75$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Computes the total integral value 86 but forgets to divide by the interval length (9 – 1) = 8.
✗ **(C)** Divides by 7 instead of 8: 86 / 7 ≈ 12.28.
✗ **(D)** Divides by the number of subintervals (4) instead of the interval width (8): 86 / 4 = 21.50.

Question 03 Analysis

★★★★ MANDATORY

Correct Answer: (C) 2 / π

STEP-BY-STEP SOLUTION

- **Average Value Formula:** $f_{avg} = (1/(\pi - 0)) \int_0^\pi \sin x\,dx$.
- **Antiderivative of Sine:** $\int \sin x\,dx = -\cos x$.
- **Evaluate the Definite Integral:**
- • $\int_0^\pi \sin x\,dx = [-\cos x]_0^\pi = (-\cos \pi) - (-\cos 0) = (-(-1)) - (-1) = 1 + 1 = 2$.
- **Compute Average Value:** $f_{avg} = (1/\pi) \cdot 2 = 2/\pi$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Confuses the interval [0, π] with a full period [0, 2π], where the integral cancels to 0.
✗ **(B)** Sign error evaluating –cos π – (–cos 0), yielding 1 instead of 2.
✗ **(D)** Evaluates the integral $\int_0^\pi \sin x\,dx = 2$ but neglects to divide by the interval length π.

Question 04 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) √3

STEP-BY-STEP SOLUTION

- **Compute the Average Value:**
- • $f_{avg} = (1/(3-0)) \int_0^3 x^2\,dx = (1/3)[x^3/3]_0^3 = (1/3)\cdot(27/3) = (1/3)\cdot 9 = 3$.
- **Set Up the MVT for Integrals Equation:**
- • The theorem guarantees a number $c \in (0, 3)$ such that $f(c) = f_{avg}$.
- • $c^2 = 3$.
- **Solve for c in the Interval:**
- • $c = \pm\sqrt{3}$. Since we require $c \in (0, 3)$, we select the positive root: $c = \sqrt{3} \approx 1.732$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Mistakes the midpoint of the domain [0, 3], which is 1.5, for the point where the average value occurs.
✗ **(B)** Algebra error setting $c^2/2 = 3$ giving $c = \sqrt{6}$.
✗ **(C)** Reports the average value itself (3) rather than the input c where f(c) = 3.

Question 05 Analysis

★★★★ MANDATORY

Correct Answer: (A) 50°F

STEP-BY-STEP SOLUTION

- **Set Up Average Value Formula:** $T_{avg} = (1/24) \int_0^{24} [50 + 10\sin(\pi t/12)]\,dt$.
- **Split the Integral:**
- • $T_{avg} = (1/24)[\int_0^{24} 50\,dt + 10\int_0^{24} \sin(\pi t/12)\,dt]$.
- **Evaluate the Constant Part:** $(1/24)\cdot(50\cdot 24) = 50$.
- **Evaluate the Sinusoidal Part:**
- • The argument $u = \pi t/12$ covers the interval $[0, 2\pi]$ as t ranges from 0 to 24.
- • The integral of sine over any full period $[0, 2\pi]$ is exactly 0.
- • $10\int_0^{24} \sin(\pi t/12)\,dt = 0$.
- **Final Result:** $T_{avg} = 50 + 0 = 50°F$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Averages the baseline 50 and maximum 60 to get 55°F.
✗ **(C)** Reports the maximum temperature $T_{max} = 50 + 10 = 60°F$.
✗ **(D)** Integrates over a half-cycle [0, 12] instead of the full 24-hour period.

Question 06 Analysis

★★★★ MANDATORY

Correct Answer: (B) 4/3

STEP-BY-STEP SOLUTION

- **Average Value Formula:** $f_{avg} = (1/(4-0)) \int_0^4 x^{(1/2)}\,dx$.
- **Compute the Definite Integral:**
- • $\int_0^4 x^{(1/2)}\,dx = [(2/3)x^{(3/2)}]_0^4$.
- • At x = 4: $(2/3)\cdot(4^{(3/2)}) = (2/3)\cdot 8 = 16/3$.
- • At x = 0: 0.
- • Value of integral = 16/3.
- **Divide by the Interval Length:** $f_{avg} = (1/4)\cdot(16/3) = 4/3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Divides 16/3 by 2 instead of the full interval width 4.
✗ **(C)** Reports the integral value 16/3 without dividing by (b – a) = 4.
✗ **(D)** Averages f(0) = 0 and f(4) = 2 linearly: (0 + 2)/2 = 1 or miscomputes.

Question 07 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 60

STEP-BY-STEP SOLUTION

- **Interpret the Average Value:**
- • $f_{avg} = (1/(5-1)) \int_1^5 f(x)\,dx = 6$.
- • $(1/4) \int_1^5 f(x)\,dx = 6 \Rightarrow \int_1^5 f(x)\,dx = 6 \cdot 4 = 24$.
- **Expand the Target Integral:**
- • $\int_1^5 (2f(x)+3)\,dx = 2\int_1^5 f(x)\,dx + \int_1^5 3\,dx$.
- **Evaluate Each Term:**
- • First term: $2 \cdot (24) = 48$.
- • Second term: $3 \cdot (5-1) = 3 \cdot 4 = 12$.
- **Combine Terms:** $48 + 12 = 60$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Substitutes the average value 6 directly without multiplying by interval width: $2(6) + 3(5) = 27$.

✗ **(B)** Forgets to add the integral of the constant 3: evaluates only $2 \cdot 18$ or 36.

✗ **(D)** Evaluates $2 \cdot 24 = 48$ but completely forgets the $\int_1^5 3\,dx$ term.

Question 08 Analysis

★★★★ MANDATORY

Correct Answer: (D) 3 / (2 ln 2)

STEP-BY-STEP SOLUTION

- **Average Value Formula:** $f_{avg} = (1/(\ln 2 - 0)) \int_0^{\ln 2} e^{2x}\,dx$.
- **Antiderivative of e^{2x}:** $\int e^{2x}\,dx = (1/2)e^{2x}$.
- **Evaluate the Definite Integral:**
- • $\int_0^{\ln 2} e^{2x}\,dx = [(1/2)e^{2x}]_0^{(\ln 2)}$.
- • At upper limit $x = \ln 2$: $(1/2)e^{2\ln 2} = (1/2)e^{\ln(2^2)} = (1/2) \cdot 4 = 2$.
- • At lower limit $x = 0$: $(1/2)e^0 = 1/2$.
- • Integral value $= 2 - 1/2 = 3/2$.
- **Compute Average Value:** $f_{avg} = (1/\ln 2) \cdot (3/2) = 3/(2 \ln 2)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Evaluates $e^{2\ln 2} = 4$ but neglects subtracting the lower limit value 1/2.

✗ **(B)** Forgets the factor of 1/2 when antidifferentiating e^{2x}, getting integral = 3 and $f_{avg} = 3/\ln 2$.

✗ **(C)** Computes the integral 3/2 but omits the factor of $1/(b-a) = 1/\ln 2$.

Question 09 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 15.33 gal/hr

STEP-BY-STEP SOLUTION

- **Formula for Average Rate:** $R_{avg} = (1/(6-0)) \int_0^6 R(t)\,dt = (1/6) \int_0^6 R(t)\,dt$.
- **Compute the Left Riemann Sum:**
- • With 3 equal subintervals on $[0, 6]$, $\Delta t = 2$.
- • Subintervals: $[0, 2]$, $[2, 4]$, $[4, 6]$ with left endpoints $t = 0, 2, 4$.
- • Left sum $= \Delta t \cdot [R(0) + R(2) + R(4)] = 2 \cdot [10 + 14 + 22] = 2 \cdot (46) = 92$ gallons.
- **Calculate Average Rate:**
- • $R_{avg} \approx (1/6) \cdot 92 = 92/6 = 46/3 \approx 15.33$ gallons/hour.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Adds the three left heights $(10 + 14 + 22 = 46)$ but omits both the width multiplier $\Delta t = 2$ and division by 6, or divides 92 by 2.

✗ **(C)** Reports total water consumed (92 gallons) instead of the average rate.

✗ **(D)** Computes the Right Riemann sum average instead: $2(14 + 22 + 18)/6 = 18.00$ gal/hr.

Question 10 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) π

STEP-BY-STEP SOLUTION

- **Geometric Interpretation of the Integral:**
- • The graph of $y = \sqrt{(16 - x^2)}$ represents the upper half of a circle centered at (0, 0) with radius r = 4.
- • The region under the curve on [−4, 4] is a semicircle of radius 4.
- **Compute the Area (Definite Integral):**
- • $\int_{-4}^{4} \sqrt{(16 - x^2)}\, dx = (1/2)\,\pi\, r^2 = (1/2) \cdot \pi \cdot (4^2) = (1/2) \cdot 16\pi = 8\pi$.
- **Find the Interval Length:** $b - a = 4 - (-4) = 8$.
- **Compute Average Value:**
- • $f_avg = (1 / 8) \cdot (8\pi) = \pi$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses radius r = 2 instead of r = 4, or divides 8π by 4 instead of 8.
✗ **(C)** Uses the full circle area 16π and divides by 4: 4π.
✗ **(D)** Reports the area of the semicircle (8π) without dividing by the interval width 8.

Question 11 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 7

STEP-BY-STEP SOLUTION

- **Fundamental Theorem of Calculus for Position:** $x(3) = x(1) + \int_{1}^{3} v(t)\, dt$.
- **Given Values:** $x(1) = 5$ and $v(t) = 3t^2 - 12$.
- **Compute the Definite Integral (Displacement):**
- • $\int_{1}^{3} (3t^2 - 12)\, dt = [\, t^3 - 12t \,]_1^3$.
- • At t = 3: $3^3 - 12(3) = 27 - 36 = -9$.
- • At t = 1: $1^3 - 12(1) = 1 - 12 = -11$.
- • Displacement = $(-9) - (-11) = -9 + 11 = 2$.
- **Determine Final Position:** $x(3) = 5 + 2 = 7$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Adds 9 instead of 2 to the initial position 5: 5 + 9 = 14.
✗ **(B)** Calculates the displacement $\int_{0}^{3} v(t)\, dt$ instead of integrating over the interval [1, 3].
✗ **(D)** Subtracts displacement 2 from 5 instead of adding: 5 − 2 = 3.

Question 12 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) 13

STEP-BY-STEP SOLUTION

- **Displacement vs. Total Distance:** Total distance is given by $\int_{0}^{5} |v(t)|\, dt$.
- **Find Turning Points:** Set $v(t) = 0 \Rightarrow 6 - 2t = 0 \Rightarrow t = 3$.
- **Analyze Signs of Velocity:**
- • On [0, 3], $v(t) \geq 0$ (moving right/forward).
- • On [3, 5], $v(t) \leq 0$ (moving left/backward).
- **Split the Integral at t = 3:**
- • Distance = $\int_{0}^{3} (6 - 2t)\, dt + \int_{3}^{5} -(6 - 2t)\, dt$.
- • First piece: $[\, 6t - t^2 \,]_0^3 = (18 - 9) - 0 = 9$.
- • Second piece: $\int_{3}^{5} (2t - 6)\, dt = [\, t^2 - 6t \,]_3^5 = (25 - 30) - (9 - 18) = -5 - (-9) = 4$.
- **Total Distance Traveled:** 9 + 4 = 13.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Computes displacement $\int_{0}^{5} (6 - 2t)\, dt = [6t - t^2]_0^5 = 30 - 25 = 5$ instead of total distance.
✗ **(B)** Considers only the forward motion from t = 0 to t = 3, ignoring the return distance.
✗ **(C)** Averages the endpoint velocities and multiplies by Δt: $((6 + (-4)) / 2) \cdot 5$, yielding an incorrect value.

Question 13 Analysis

★★★★ MANDATORY

Correct Answer: (A) 21.5 m

STEP-BY-STEP SOLUTION

- **Definition of Displacement:** Displacement = $\int_0^{10} v(t)\,dt$.
- **Compute Trapezoidal Areas with Variable Widths Δt:**
- • Interval $[0, 2]$: $\Delta t = 2$. $\text{Area}_1 = (2/2) \cdot (3 + 7) = 1 \cdot 10 = 10$.
- • Interval $[2, 5]$: $\Delta t = 3$. $\text{Area}_2 = (3/2) \cdot (7 + 4) = 1.5 \cdot 11 = 16.5$.
- • Interval $[5, 8]$: $\Delta t = 3$. $\text{Area}_3 = (3/2) \cdot (4 + (-2)) = 1.5 \cdot 2 = 3.0$.
- • Interval $[8, 10]$: $\Delta t = 2$. $\text{Area}_4 = (2/2) \cdot (-2 + (-6)) = 1 \cdot (-8) = -8.0$.
- **Sum the Trapezoids for Net Displacement:**
- • Net Displacement $\approx 10 + 16.5 + 3.0 + (-8.0) = 21.5$ m.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Miscalculates Area_2 as 11.5 or omits interval $[5, 8]$, resulting in 16.5 m.

✗ **(C)** Takes the absolute value of all terms (approximating total distance instead of displacement): $10 + 16.5 + 3 + 8 = 37.5$ m, or calculation error yielding 31.5 m.

✗ **(D)** Assumes uniform width $\Delta t = 2.5$ across all subintervals.

Question 14 Analysis

★★★★ MANDATORY

Correct Answer: (B) $2e^4 + 1$

STEP-BY-STEP SOLUTION

- **Velocity from Acceleration:** $v(t) = v(0) + \int_0^t a(u)\,du$.
- **Evaluate the Indefinite Integral:**
- • $\int 4e^{2u}\,du = 2e^{2u} + C$.
- **Apply the Definite Integral from 0 to 2:**
- • $\int_0^2 4e^{2u}\,du = [\,2e^{2u}\,]_0^2 = 2e^4 - 2e^0 = 2e^4 - 2$.
- **Add Initial Velocity v(0) = 3:**
- • $v(2) = 3 + (2e^4 - 2) = 2e^4 + 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to subtract the lower limit evaluation $2e^0 = 2$, adding 3 directly to $2e^4$.

✗ **(C)** Forgets the 1/2 factor from u-substitution on e^{2t}, integrating to $4e^{2t}$.

✗ **(D)** Multiplies by 2 instead of dividing by 2 when integrating.

Question 15 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) $6 / \pi$

STEP-BY-STEP SOLUTION

- **Total Distance Formula:** Distance = $\int_0^3 |\cos(\pi t / 2)|\,dt$.
- **Find Turning Points in [0, 3]:**
- • $\cos(\pi t / 2) = 0 \Rightarrow \pi t / 2 = \pi/2, 3\pi/2 \Rightarrow t = 1, t = 3$.
- **Determine Signs of v(t):**
- • On $[0, 1]$: $\cos(\pi t / 2) \geq 0$.
- • On $[1, 3]$: $\cos(\pi t / 2) \leq 0$.
- **Split the Integral:**
- • Distance = $\int_0^1 \cos(\pi t / 2)\,dt + \int_1^3 -\cos(\pi t / 2)\,dt$.
- • Part 1: $[\,(2/\pi)\sin(\pi t / 2)\,]_0^1 = (2/\pi)(\sin(\pi/2) - \sin(0)) = 2/\pi$.
- • Part 2: $-[\,(2/\pi)\sin(\pi t / 2)\,]_1^3 = -(2/\pi)(\sin(3\pi/2) - \sin(\pi/2)) = -(2/\pi)(-1 - 1) = 4/\pi$.
- **Total Distance:** $2/\pi + 4/\pi = 6/\pi$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Calculates net displacement: $\int_0^3 \cos(\pi t/2)\,dt = [(2/\pi)\sin(\pi t/2)]_0^3 = (2/\pi)(-1 - 0) = -2/\pi$, taking magnitude $2/\pi$.

✗ **(B)** Calculates only the distance traveled during the second interval $[1, 3]$ which is $4/\pi$.

✗ **(D)** Assumes periodic symmetry cancels total distance to zero.

Question 16 Analysis

★★★★ MANDATORY

Correct Answer: (D) √41 – 6

STEP-BY-STEP SOLUTION

- **Position from Velocity:** $s(6) = s(2) + \int_2^6 (t / \sqrt{(t^2 + 5)})\, dt$.
- **Evaluate the Integral using Substitution:**
- • Let $u = t^2 + 5 \Rightarrow du = 2t\, dt \Rightarrow t\, dt = du / 2$.
- • When $t = 2$: $u = 2^2 + 5 = 9$.
- • When $t = 6$: $u = 6^2 + 5 = 41$.
- • $\int_9^{41} (1 / (2\sqrt{u}))\, du = [\sqrt{u}]_9^{41} = \sqrt{41} - \sqrt{9} = \sqrt{41} - 3$.
- **Add Initial Position s(2) = –3:**
- • $s(6) = -3 + (\sqrt{41} - 3) = \sqrt{41} - 6$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Adds displacement to +3 instead of –3: $3 + (\sqrt{41} - 3) = \sqrt{41}$.
✗ **(B)** Forgets the initial position $s(2) = -3$ and writes only the displacement $\sqrt{41} - 3$.
✗ **(C)** Forgets to subtract $\sqrt{9} = 3$ at the lower limit and forgets initial position.

Question 17 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 10 – 2π

STEP-BY-STEP SOLUTION

- **Position Formula:** $x(8) = x(0) + \int_0^8 v(t)\, dt$.
- **Break Area into Geometric Components:**
- • Triangle from $t = 0$ to $t = 4$: base = 4, height = 4.
- • Area of triangle = $(1/2) \cdot 4 \cdot 4 = 8$.
- • Region from $t = 4$ to $t = 8$: Semicircle lying below the t-axis.
- • Diameter = $8 - 4 = 4$, so radius $r = 2$.
- • Area of semicircle = $(1/2) \cdot \pi \cdot r^2 = (1/2) \cdot \pi \cdot (2)^2 = 2\pi$.
- • Since it lies below the axis, $\int_4^8 v(t)\, dt = -2\pi$.
- **Net Displacement:** $\int_0^8 v(t)\, dt = 8 - 2\pi$.
- **Position at t = 8:** $x(8) = 2 + (8 - 2\pi) = 10 - 2\pi$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses radius $r = 4$ instead of $r = 2$ for the semicircle, obtaining area $8\pi / 2 = 4\pi$.
✗ **(C)** Forgets to add the initial position $x(0) = 2$, giving only the displacement $8 - 2\pi$.
✗ **(D)** Treats the semicircle as positive area above the t-axis: $2 + 8 + 2\pi = 10 + 2\pi$.

Question 18 Analysis

★★★ IMPORTANT

Correct Answer: (B) 0 < t < 2

STEP-BY-STEP SOLUTION

- **Condition for Decreasing Speed:** Speed $|v(t)|$ decreases when velocity $v(t)$ and acceleration $a(t)$ have opposite signs ($v(t) \cdot a(t) < 0$).
- **Compute Acceleration:** $a(t) = v'(t) = 4 > 0$ for all t.
- **Analyze Sign of Velocity:**
- • $4t - 8 < 0 \Rightarrow 4t < 8 \Rightarrow t < 2$.
- • So for $0 < t < 2$, $v(t) < 0$ while $a(t) > 0$.
- **Conclusion:** Since velocity and acceleration have opposite signs on (0, 2), the particle's speed is decreasing on $0 < t < 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Identifies where speed is increasing (both $v(t) > 0$ and $a(t) > 0$ for $t > 2$).
✗ **(C)** Confuses decreasing velocity with decreasing speed.
✗ **(D)** Believes constant positive acceleration implies speed is always increasing.

Question 19 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 11

STEP-BY-STEP SOLUTION

- **Find Velocity Function v(t):**
- • $v(t) = \int (12t - 4)\,dt = 6t^2 - 4t + C_1$.
- • Use condition $v(1) = 3$: $6(1)^2 - 4(1) + C_1 = 3 \Rightarrow 2 + C_1 = 3 \Rightarrow C_1 = 1$.
- • Thus, $v(t) = 6t^2 - 4t + 1$.
- **Find Position Function x(t):**
- • $x(t) = \int (6t^2 - 4t + 1)\,dt = 2t^3 - 2t^2 + t + C_2$.
- • Use condition $x(1) = 2$: $2(1)^3 - 2(1)^2 + 1 + C_2 = 2 \Rightarrow 1 + C_2 = 2 \Rightarrow C_2 = 1$.
- • Thus, $x(t) = 2t^3 - 2t^2 + t + 1$.
- **Evaluate Position at t = 2:**
- • $x(2) = 2(2)^3 - 2(2)^2 + 2 + 1 = 2(8) - 2(4) + 2 + 1 = 16 - 8 + 3 = 11$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Arithmetic error in evaluating $2(8) - 2(4) + 2 + 1$, arriving at 13.

✗ **(B)** Forgets constant $C_2 = 1$, obtaining $x(2) = 10 - 1 = 9$.

✗ **(D)** Integrates acceleration directly twice without solving constants of integration.

Question 20 Analysis

★★★★ MANDATORY

Correct Answer: (D) 1

STEP-BY-STEP SOLUTION

- **Relate Displacement to Velocity:** Displacement = $\int_0^4 v(t)\,dt = \int_0^4 k \cdot t\,dt$.
- **Evaluate Integral:** $[(k/2)\,t^2]_0^4 = (k/2)(16) = 8k$.
- **Given Displacement is 8:** $8k = 8 \Rightarrow k = 1$.
- **Verify Consistency with Total Distance:**
- • If $k = 1$, $v(t) = t \geq 0$ for all t in $[0, 4]$.
- • Since velocity never changes sign, Total Distance = Displacement = 8 meters.
- **Conclusion:** $k = 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Equates $k \cdot 4 = 8$, treating velocity as constant rather than linear.

✗ **(B)** Solves $4k = 8$ forgetting the 1/2 factor in antiderivative ($t^2/2$).

✗ **(C)** Divides 8 by 16 instead of equating $8k = 8$.

Question 21 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 300 liters

STEP-BY-STEP SOLUTION

- **Net Change Formulation:** The amount of water $W(t)$ at time t is given by $W(t) = W(0) + \int_0^t [R(u) - L(u)]\,du$.
- **Identify Net Rate of Change:**
- • Inflow rate: $R(t) = 60 - 4t$.
- • Outflow rate: $L(t) = 20$.
- • Net rate: $R(t) - L(t) = (60 - 4t) - 20 = 40 - 4t$.
- **Compute the Net Change from t = 0 to t = 5:**
- • $\int_0^5 (40 - 4u)\,du = [40u - 2u^2]_0^5$.
- • At $u = 5$: $40(5) - 2(5)^2 = 200 - 2(25) = 200 - 50 = 150$ liters.
- **Add Initial Volume:** $W(5) = 150 + 150 = 300$ liters.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets to subtract the 20 L/min outflow rate: $150 + \int_0^5 (60 - 4t)\,dt = 150 + 250 = 400$, or calculation error yielding 250.

✗ **(C)** Adds only total inflow (250 L) without subtracting outflow (100 L): $150 + 250 = 400$, or miscalculates net change as 200.

✗ **(D)** Calculates net change 150 but forgets to add the initial volume $W(0) = 150$, or subtracts net change.

Question 22 Analysis

★★★★ MANDATORY

Correct Answer: (B) 153 liters

STEP-BY-STEP SOLUTION

- **Total Leakage Formula:** Total leakage = $\int_0^{12} R(t)\,dt$.
- **Left Riemann Sum with Variable Widths Δt:**
- • Subinterval [0, 2]: $\Delta t_1 = 2 - 0 = 2$. Left value = R(0) = 18. Product = $2 \cdot 18 = 36$.
- • Subinterval [2, 5]: $\Delta t_2 = 5 - 2 = 3$. Left value = R(2) = 15. Product = $3 \cdot 15 = 45$.
- • Subinterval [5, 9]: $\Delta t_3 = 9 - 5 = 4$. Left value = R(5) = 12. Product = $4 \cdot 12 = 48$.
- • Subinterval [9, 12]: $\Delta t_4 = 12 - 9 = 3$. Left value = R(9) = 8. Product = $3 \cdot 8 = 24$.
- **Total Estimated Leakage:**
- • 36 + 45 + 48 + 24 = 153 liters.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Calculation error summing terms or miscalculating Δt values.
✗ **(C)** Uses a Right Riemann sum: 2(15) + 3(12) + 4(8) + 3(5) = 30 + 36 + 32 + 15 = 113 liters.
✗ **(D)** Assumes equal subinterval widths of 3 hours: 3(18 + 15 + 12 + 8) = 159.

Question 23 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) t = 16/3 hours

STEP-BY-STEP SOLUTION

- **Accumulation Function:** Let N(t) be the number of people in the park: $N(t) = \int_0^t [E(u) - L(u)]\,du$.
- **Find Critical Points:** By the Fundamental Theorem of Calculus, $N'(t) = E(t) - L(t)$.
- • Set $N'(t) = 0 \Rightarrow E(t) = L(t)$.
- • $120t - 15t^2 = 40t$.
- • $80t - 15t^2 = 0 \Rightarrow 5t(16 - 3t) = 0$.
- • Since $t > 0$, $16 - 3t = 0 \Rightarrow t = 16/3$ hours.
- **Verify Maximum with First Derivative Test:**
- • For $0 < t < 16/3$: $N'(t) > 0$ (park population is increasing).
- • For $16/3 < t \le 8$: $N'(t) < 0$ (park population is decreasing).
- **Conclusion:** The maximum occurs at t = 16/3 hours.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Finds the peak of the entrance rate $E'(t) = 0$: $120 - 30t = 0 \Rightarrow t = 4$, confusing peak entrance rate with maximum park occupancy.
✗ **(B)** Algebra error: solves $40t - 15t^2 = 0$ yielding $t = 8/3$.
✗ **(D)** Chooses the end of the day, assuming accumulation continues indefinitely.

Question 24 Analysis

★★★★ MANDATORY

Correct Answer: (D) 12 kg

STEP-BY-STEP SOLUTION

- **Total Amount Removed:** Total = $\int_0^6 (80/(t+4)^2)\,dt$.
- **Antiderivative:**
- • $\int 80(t+4)^{-2}\,dt = -80/(t+4)$.
- **Evaluate the Definite Integral from 0 to 6:**
- • $[-80/(t+4)]_0^6 = (-80/10) - (-80/4) = -8 - (-20) = -8 + 20 = 12$ kg.
- **Conclusion:** 12 kg of pollutant is removed.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Arithmetic error in subtracting $-8 - (-20) = -8 + 24 = 16$.
✗ **(B)** Evaluates only the lower limit term 80 / 4 = 20 kg, omitting the upper limit.
✗ **(C)** Evaluates only the magnitude of the upper limit term 80 / 10 = 8 kg.

Question 25 Analysis

★★★ IMPORTANT

Correct Answer: (A) 120 + 144 / π MW·hr

STEP-BY-STEP SOLUTION

- **Total Consumption Formula:** Total = $\int_0^{12} C(t)\,dt = \int_0^{12} [10 + 6\sin(\pi t / 12)]\,dt$.
- **Evaluate the Integral:**
- • Term 1: $\int_0^{12} 10\,dt = 10 \cdot 12 = 120$.
- • Term 2: $\int_0^{12} 6\sin(\pi t / 12)\,dt = [-6 \cdot (12/\pi)\cos(\pi t / 12)]_0^{12} = [-(72/\pi)\cos(\pi t / 12)]_0^{12}$.
- • At t = 12: $-(72/\pi)\cos(\pi) = -(72/\pi)(-1) = 72/\pi$.
- • At t = 0: $-(72/\pi)\cos(0) = -(72/\pi)(1) = -72/\pi$.
- • Integral of Term 2 = $(72/\pi) - (-72/\pi) = 144/\pi$.
- **Combine Terms:** Total = $120 + 144/\pi$ MW·hr.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Evaluates $[-(72/\pi)\cos(\pi t/12)]_0^{12}$ with sign error as $72/\pi - 0 = 72/\pi$.
✗ **(C)** Believes the integral of sine over a half period cancels to 0 (confusing it with a full period [0, 24]).
✗ **(D)** Sign error in integrating sine, writing $-144/\pi$ instead of $+144/\pi$.

Question 26 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 1,500 e – 300

STEP-BY-STEP SOLUTION

- **Net Change for Population:** $P(5) = P(0) + \int_0^5 R(t)\,dt$.
- **Given Initial Condition:** P(0) = 1,200.
- **Compute the Integral:**
- • $\int_0^5 300\,e^{0.2t}\,dt = [300 / 0.2 \cdot e^{0.2t}]_0^5 = [1{,}500\,e^{0.2t}]_0^5$.
- • At t = 5: $1{,}500\,e^{0.2 \cdot 5} = 1{,}500\,e^1 = 1{,}500\,e$.
- • At t = 0: $1{,}500\,e^0 = 1{,}500$.
- • Total net increase = 1,500 e – 1,500.
- **Add Initial Population:**
- • P(5) = 1,200 + (1,500 e – 1,500) = 1,500 e – 300.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to subtract lower limit 1,500 and forgets to add initial 1,200.
✗ **(C)** Reports only the net increase $\int_0^5 R(t)\,dt = 1{,}500\,e - 1{,}500$, omitting initial population.
✗ **(D)** Multiplies initial population by e directly: 1,200 e.

Question 27 Analysis

★★★★ MANDATORY

Correct Answer: (C) 33°C

STEP-BY-STEP SOLUTION

- **Net Change in Temperature:** $T(6) = T(0) + \int_0^6 [H(t) - W(t)]\,dt$.
- **Compute the Integral:**
- • $\int_0^6 (4 - 0.5t)\,dt = [4t - 0.25t^2]_0^6$.
- • At t = 6: $4(6) - 0.25(6)^2 = 24 - 0.25(36) = 24 - 9 = 15°C$.
- **Add Initial Temperature T(0) = 18°C:**
- • T(6) = 18 + 15 = 33°C.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Multiplies 0.5 by 36 instead of 0.25: 24 – 18 = 6, adding 6 to 18 gives 24, or arithmetic error.
✗ **(B)** Reports only the net temperature increase (15°C) without adding the initial 18°C.
✗ **(D)** Subtracts 3 instead of 9 in evaluating 24 – 9 = 15.

Question 28 Analysis

★★★★ MANDATORY

Correct Answer: (D) 76 tons

STEP-BY-STEP SOLUTION

- **Total Gravel Added:** Total = $\int_1^3 (12 + 6t^2)\,dt$.
- **Find Antiderivative:** $[12t + 2t^3]_1^3$.
- **Evaluate at Limits:**
- • At $t = 3$: $12(3) + 2(3)^3 = 36 + 2(27) = 36 + 54 = 90$.
- • At $t = 1$: $12(1) + 2(1)^3 = 12 + 2 = 14$.
- **Subtract Limits:** $90 - 14 = 76$ tons.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Integrates $6t^2$ to $3t^3$ instead of $2t^3$.
✗ **(B)** Calculates $90 - 26 = 64$ due to arithmetic error evaluating at $t = 1$.
✗ **(C)** Integrates from 0 to 3 instead of 1 to 3: $[12t + 2t^3]_0^3 = 90$ tons, minus 10 = 80.

Question 29 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 17.33 kg

STEP-BY-STEP SOLUTION

- **Mass as an Integral of Linear Density:** Total mass $M = \int_0^4 \rho(x)\,dx = \int_0^4 (3 + 2x - 0.5x^2)\,dx$.
- **Find Antiderivative:**
- • $\int (3 + 2x - 0.5x^2)\,dx = [3x + x^2 - (1/6)x^3]_0^4$.
- **Evaluate at x = 4:**
- • $3(4) + (4)^2 - (1/6)(4)^3 = 12 + 16 - (1/6)(64)$.
- • $28 - 64/6 = 28 - 32/3 = 84/3 - 32/3 = 52/3$ kg.
- • In decimal form: $52 / 3 = 17.33$ kg.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Omit linear term or neglects quadratic term: $12 + 16 - 12 = 16.00$ kg.
✗ **(C)** Adds rather than subtracts the cubic term: $28 + 10.67 = 38.67$ kg.
✗ **(D)** Approximates by averaging endpoint densities: $\rho(0) = 3$, $\rho(4) = 3 + 8 - 8 = 3$, yielding 12 kg.

Question 30 Analysis

★★★★ MANDATORY

Correct Answer: (B) 270 people

STEP-BY-STEP SOLUTION

- **Accumulation as Area Under the Rate Curve:** Total people = $\int_0^6 E(t)\,dt$.
- **Decompose the Area Under E(t):**
- • Region 1: Rectangle from $t = 0$ to $t = 3$ with height 60.
- • $\text{Area}_1 = \text{width} \cdot \text{height} = 3 \cdot 60 = 180$ people.
- • Region 2: Triangle from $t = 3$ to $t = 6$ with base = 3 and height = 60.
- • $\text{Area}_2 = (1/2) \cdot \text{base} \cdot \text{height} = (1/2) \cdot 3 \cdot 60 = 90$ people.
- **Sum the Geometric Areas:**
- • Total = 180 + 90 = 270 people.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Calculates area of a rectangle across the entire 6–hour interval: $6 \cdot 60 = 360$ people.
✗ **(C)** Accounts only for the first 3 hours (rectangle area = 180 people).
✗ **(D)** Uses average rate $(60 + 0)/2 = 30$ across 6 hours or calculation error yielding 240.

Question 31 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 4 / 3

STEP-BY-STEP SOLUTION

- **Find Points of Intersection:** Set $2x = x^2 \Rightarrow x^2 - 2x = 0 \Rightarrow x(x-2) = 0 \Rightarrow x = 0$ and $x = 2$.
- **Identify Top and Bottom Functions on [0, 2]:** For $0 < x < 2$, $2x > x^2$, so $y_{top} = 2x$ and $y_{bottom} = x^2$.
- **Set Up the Area Integral:** Area = $\int_0^2 (2x - x^2)\,dx$.
- **Evaluate the Definite Integral:**
- • $[x^2 - (1/3)x^3]_0^2$.
- • At $x = 2$: $(2)^2 - (1/3)(2)^3 = 4 - 8/3 = 12/3 - 8/3 = 4/3$.
- • At $x = 0$: $0 - 0 = 0$.
- **Conclusion:** Area = 4/3.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Approximates area as a triangle with base 2 and height 2, yielding 2.
✗ **(B)** Forgets to square x in the first term: $2 - 4/3 = 2/3$.
✗ **(D)** Adds the two functions instead of subtracting: $\int_0^2 (2x + x^2)\,dx = 4 + 8/3 = 20/3$ or arithmetic mistake.

Question 32 Analysis

★★★★ MANDATORY

Correct Answer: (D) √2 – 1

STEP-BY-STEP SOLUTION

- **Identify Relative Positions of Curves:** On $[0, \pi/4]$, $\cos x \geq \sin x$, so $y_{top} = \cos x$ and $y_{bottom} = \sin x$.
- **Set Up the Integral:** Area = $\int_0^{\pi/4} (\cos x - \sin x)\,dx$.
- **Antiderivative:**
- • $\int (\cos x - \sin x)\,dx = \sin x - (-\cos x) = \sin x + \cos x$.
- **Evaluate at Limits:**
- • At $x = \pi/4$: $\sin(\pi/4) + \cos(\pi/4) = \sqrt{2}/2 + \sqrt{2}/2 = \sqrt{2}$.
- • At $x = 0$: $\sin(0) + \cos(0) = 0 + 1 = 1$.
- **Compute Net Area:** Area = $\sqrt{2} - 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Arithmetic error in adding $\sqrt{2}/2 + \sqrt{2}/2$, writing 2.
✗ **(B)** Neglects to subtract the evaluation at lower limit $x = 0$.
✗ **(C)** Reverses order of subtraction: $1 - \sqrt{2}/2$.

Question 33 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 9 / 2

STEP-BY-STEP SOLUTION

- **Find Points of Intersection:**
- • $4 - x^2 = x + 2 \Rightarrow x^2 + x - 2 = 0$.
- • $(x+2)(x-1) = 0 \Rightarrow x = -2$ and $x = 1$.
- **Determine Relative Position:** For $-2 < x < 1$, $4 - x^2 > x + 2$.
- **Set Up the Area Integral:**
- • Area = $\int_{-2}^{1} [(4 - x^2) - (x + 2)]\,dx = \int_{-2}^{1} (2 - x - x^2)\,dx$.
- **Find Antiderivative:**
- • $[2x - (1/2)x^2 - (1/3)x^3]_{-2}^{1}$.
- **Evaluate at Limits:**
- • At $x = 1$: $2(1) - 1/2 - 1/3 = 2 - 5/6 = 7/6$.
- • At $x = -2$: $2(-2) - (1/2)(4) - (1/3)(-8) = -4 - 2 + 8/3 = -6 + 8/3 = -10/3$.
- **Compute Difference:** $7/6 - (-10/3) = 7/6 + 20/6 = 27/6 = 9/2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Arithmetic error in combining fractions with common denominator 6, getting $21/6 = 7/2$.
✗ **(C)** Integrates parabola alone over symmetric interval [0, 2]: 16/3.
✗ **(D)** Forgets the 1/2 factor in antiderivative of linear term x.

Question 34 Analysis

★★★★ MANDATORY

Correct Answer: (B) 4 / 3

STEP-BY-STEP SOLUTION

- **Compare Functions on [0, ln 3]:** For $x \geq 0$, $e^x \geq e^{-x}$, so y_top = e^x and y_bottom = e^{-x}.
- **Set Up Integral:** Area = $\int_0^{\ln 3} (e^x - e^{-x})\,dx$.
- **Antiderivative:** $[e^x - (-e^{-x})] = [e^x + e^{-x}]_0^{(\ln 3)}$.
- **Evaluate at Upper Limit x = ln 3:**
- • $e^{\ln 3} + e^{-\ln 3} = 3 + 1/3 = 10/3$.
- **Evaluate at Lower Limit x = 0:**
- • $e^0 + e^0 = 1 + 1 = 2$.
- **Compute Difference:** $10/3 - 2 = 10/3 - 6/3 = 4/3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Evaluates lower limit as 0 instead of 2, subtracting 2/3 instead of 2: $10/3 - 2/3 = 8/3$.
✗ **(C)** Approximates area or calculates $e^{\ln 3} - 1 = 2$.
✗ **(D)** Reports upper limit value 10/3 without subtracting lower limit evaluation 2.

Question 35 Analysis

★★★★ MANDATORY

Correct Answer: (C) 4 / 3

STEP-BY-STEP SOLUTION

- **Find Intersections:** $\sqrt{x} = x/2 \Rightarrow x = x^2/4 \Rightarrow x^2 - 4x = 0 \Rightarrow x = 0$ and $x = 4$.
- **Relative Position:** On (0, 4), $\sqrt{x} > x/2$.
- **Set Up the Area Integral:** Area = $\int_0^4 (x^{(1/2)} - x/2)\,dx$.
- **Evaluate the Definite Integral:**
- • $[(2/3)x^{(3/2)} - x^2/4]_0^4$.
- • At x = 4: $(2/3)(4)^{(3/2)} - (4)^2/4 = (2/3)(8) - 16/4 = 16/3 - 4 = 16/3 - 12/3 = 4/3$.
- • At x = 0: $0 - 0 = 0$.
- **Conclusion:** Area = 4/3.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to subtract the area under y = x/2: reports only (2/3)(8) = 16/3 or miscalculates to 8/3.
✗ **(B)** Uses coefficient 1/3 instead of 2/3 for antiderivative of $x^{(1/2)}$: $8/3 - 4 = -4/3$.
✗ **(D)** Uses a linear approximation triangle with base 4 and height 1, getting 2.

Question 36 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) 4

STEP-BY-STEP SOLUTION

- **Find Points of Intersection in First Quadrant (x ≥ 0):**
- • $x^3 - 3x = x \Rightarrow x^3 - 4x = 0 \Rightarrow x(x^2 - 4) = 0$.
- • In the first quadrant, x = 0 and x = 2.
- **Identify Top and Bottom:** For $0 < x < 2$, test x = 1: y = 1 vs $y = 1^3 - 3(1) = -2$. Thus y_top = x and y_bottom = $x^3 - 3x$.
- **Set Up Area Integral:**
- • Area = $\int_0^2 [x - (x^3 - 3x)]\,dx = \int_0^2 (4x - x^3)\,dx$.
- **Evaluate the Integral:**
- • $[2x^2 - (1/4)x^4]_0^2$.
- • At x = 2: $2(2)^2 - (1/4)(2)^4 = 2(4) - (1/4)(16) = 8 - 4 = 4$.
- • At x = 0: 0.
- **Conclusion:** Area = 4.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Integrates $4x - x^3$ with antiderivative $4x^2 - x^4$: $16 - 16 = 0$, or arithmetic mistake giving 6.
✗ **(B)** Integrates only to x = 1 or halves the result.
✗ **(C)** Includes the symmetric region in the third quadrant (from x = –2 to 0), failing to restrict to the first quadrant.

Question 37 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 4^(1/3)

STEP-BY-STEP SOLUTION

- **Total Area of the Region:** Total = $\int_0^2 x^2\,dx = [x^3/3]_0^2 = 8/3$.
- **Condition for Equal Subregions:** The area from $x = 0$ to $x = k$ must equal half of the total area.
- • Half Area = $(1/2)\cdot(8/3) = 4/3$.
- **Set Up Integral Equation for k:**
- • $\int_0^k x^2\,dx = 4/3$.
- • $[x^3/3]_0^k = k^3/3 = 4/3$.
- **Solve for k:** $k^3 = 4 \Rightarrow k = 4^{(1/3)} = \sqrt[3]{4}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Divides 8/3 by 4 instead of 2, getting $k^3 = 2 \Rightarrow k = 2^{(1/3)}$.

✗ **(C)** Confuses equal area with the Mean Value Theorem for Integrals ($x^2 = 4/3 \Rightarrow x = 2/\sqrt{3}$).

✗ **(D)** Assumes the midpoint of the interval [0, 2] splits the area equally, ignoring the curvature of x^2.

Question 38 Analysis

★★★ IMPORTANT

Correct Answer: (B) 2

STEP-BY-STEP SOLUTION

- **Area Formulation:** Area = $\int_1^{e^2} (1/x)\,dx$.
- **Evaluate Antiderivative:** $[\ln|x|]_1^{(e^2)}$.
- **Evaluate at Limits:**
- • At $x = e^2$: $\ln(e^2) = 2$.
- • At $x = 1$: $\ln(1) = 0$.
- **Compute Difference:** $2 - 0 = 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Calculates $\ln(e) = 1$ instead of $\ln(e^2) = 2$.

✗ **(C)** Treats 1/x as a polynomial and uses power rule: $[x^0]_1^{(e^2)} = e^2 - 1$.

✗ **(D)** Multiplies exponent by base: $2e$.

Question 39 Analysis

★★★★ MANDATORY

Correct Answer: (C) 32 / 3

STEP-BY-STEP SOLUTION

- **Find Points of Intersection:** $3 - x^2 = -1 \Rightarrow x^2 = 4 \Rightarrow x = -2$ and $x = 2$.
- **Identify Top and Bottom Curves:** For $-2 < x < 2$, $3 - x^2 > -1$.
- • Top: $y = 3 - x^2$.
- • Bottom: $y = -1$.
- **Set Up the Area Integral:**
- • Area = $\int_{-2}^{2} [(3 - x^2) - (-1)]\,dx = \int_{-2}^{2} (4 - x^2)\,dx$.
- **Use Symmetry:**
- • Area = $2\int_0^2 (4 - x^2)\,dx = 2[4x - (1/3)x^3]_0^2$.
- • At $x = 2$: $4(2) - (1/3)(8) = 8 - 8/3 = 16/3$.
- • Multiply by 2: $2\cdot(16/3) = 32/3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Subtracts 1 instead of −1, integrating $2 - x^2$ over $[-2, 2]$.

✗ **(B)** Calculates only the right half from $x = 0$ to $x = 2$ (16/3), forgetting to multiply by 2.

✗ **(D)** Doubles the correct area: 64/3.

Question 40 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) 44

STEP-BY-STEP SOLUTION

- **Area Formulation:** Area = $\int_0^6 [f(x) - g(x)]\,dx$.
- **Compute the Difference h(x) = f(x) – g(x) at Each Point:**
- • At x = 0: h(0) = 10 – 2 = 8.
- • At x = 2: h(2) = 14 – 4 = 10.
- • At x = 4: h(4) = 11 – 5 = 6.
- • At x = 6: h(6) = 8 – 4 = 4.
- **Apply Trapezoidal Rule with Equal Width Δx = 2:**
- • Area ≈ (Δx / 2) · [h(0) + 2h(2) + 2h(4) + h(6)].
- • (2 / 2) · [8 + 2(10) + 2(6) + 4] = 1 · [8 + 20 + 12 + 4] = 44.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Divides by interval width 6 or 2, calculating an average value.

✗ **(B)** Uses a Right Riemann sum: 2(10 + 6 + 4) = 40 or arithmetic mistake.

✗ **(C)** Uses a Left Riemann sum: 2(8 + 10 + 6) = 48 or miscounts 50.

Question 41 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 9 / 2

STEP-BY-STEP SOLUTION

- **Identify Limits of Integration in y:**
- • Set $y^2 = y + 2 \Rightarrow y^2 - y - 2 = 0$.
- • $(y - 2)(y + 1) = 0 \Rightarrow y = -1$ and $y = 2$.
- **Horizontal Slicing (Right – Left):**
- • On the interval [–1, 2], the line lies to the right of the parabola.
- • $x_{right} = y + 2$, and $x_{left} = y^2$.
- **Set Up the Area Integral:**
- • Area = $\int_{-1}^{2} [(y + 2) - y^2]\,dy = \int_{-1}^{2} (2 + y - y^2)\,dy$.
- **Evaluate the Definite Integral:**
- • $[2y + (1/2)y^2 - (1/3)y^3]_{-1}^{2}$.
- • At y = 2: 2(2) + (1/2)(4) – (1/3)(8) = 4 + 2 – 8/3 = 6 – 8/3 = 10/3.
- • At y = –1: 2(–1) + (1/2)(1) – (1/3)(–1) = –2 + 1/2 + 1/3 = –2 + 5/6 = –7/6.
- • Difference: 10/3 – (–7/6) = 20/6 + 7/6 = 27/6 = 9/2.

WHY OTHER OPTIONS ARE INCORRECT

Question 42 Analysis

★★★★ MANDATORY

Correct Answer: (B) 32 / 3

STEP-BY-STEP SOLUTION

- **Find y-intercepts:** Set $4 - y^2 = 0 \Rightarrow y = -2$ and $y = 2$.
- **Identify Boundaries:** For $-2 \le y \le 2$, $x_{right} = 4 - y^2$ and $x_{left} = 0$.
- **Set Up Integral:** Area = $\int_{-2}^{2} (4 - y^2)\,dy$.
- **Use Symmetry Across y = 0:**
- • Area = $2\int_0^2 (4 - y^2)\,dy = 2[4y - (1/3)y^3]_0^2$.
- • At y = 2: 4(2) – (1/3)(8) = 8 – 8/3 = 16/3.
- • Multiply by 2: 2 · (16/3) = 32/3.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Integrates only over [0, 2], forgetting to double for the region below the x-axis: 16/3.

✗ **(C)** Forgets factor of 4: 8/3.

✗ **(D)** Doubles the correct total area: 64/3.

Question 43 Analysis

★★★★ MANDATORY

Correct Answer: (C) $e^2 - 3$

STEP-BY-STEP SOLUTION

- **Determine Integration Boundaries in y:**
- • Intersection of $x = e^y$ and $x = 1$: $e^y = 1 \Rightarrow y = 0$.
- • Top horizontal boundary: $y = 2$.
- • So y varies from $y = 0$ to $y = 2$.
- **Horizontal Slicing:** $x_{right} = e^y$ and $x_{left} = 1$.
- **Set Up the Area Integral:** Area = $\int_0^2 (e^y - 1)\,dy$.
- **Evaluate Antiderivative:** $[e^y - y]_0^2$.
- • At $y = 2$: $e^2 - 2$.
- • At $y = 0$: $e^0 - 0 = 1$.
- **Compute Difference:** $(e^2 - 2) - 1 = e^2 - 3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Neglects to subtract the lower limit evaluation $e^0 = 1$.

✗ **(B)** Forgets to integrate the constant function 1, evaluating $\int_0^2 e^y\,dy = e^2 - 1$.

✗ **(D)** Multiplies by 2 instead of integrating over the interval.

Question 44 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) 1 / 2

STEP-BY-STEP SOLUTION

- **Understand Total Area vs. Net Integral:**
- • The net integral $\int_{-1}^{1} (x^3 - x)\,dx = 0$ because the function is odd.
- • Total area requires taking the absolute value: Area = $\int_{-1}^{1} |x^3 - x|\,dx$.
- **Find Intersections:** $x^3 - x = 0 \Rightarrow x(x^2 - 1) = 0 \Rightarrow x = -1, 0, 1$.
- **Analyze Signs and Use Symmetry:**
- • On $[-1, 0]$, $x^3 - x \geq 0$.
- • On $[0, 1]$, $x^3 - x \leq 0$ (so top curve is $y = 0$ and bottom is $y = x^3 - x$).
- • By symmetry, Area = $2\int_0^1 [0 - (x^3 - x)]\,dx = 2\int_0^1 (x - x^3)\,dx$.
- **Evaluate the Definite Integral:**
- • $2[(1/2)x^2 - (1/4)x^4]_0^1 = 2[1/2 - 1/4] = 2 \cdot (1/4) = 1/2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Doubles the correct area: 1.

✗ **(B)** Computes the net integral $\int_{-1}^{1} (x^3 - x)\,dx = 0$ instead of total area.

✗ **(C)** Calculates the area of only one loop $[0, 1]$ without doubling for symmetry: 1/4.

Question 45 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $2\sqrt{2}$

STEP-BY-STEP SOLUTION

- **Find Point of Intersection in $[0, \pi]$:** $\sin x = \cos x \Rightarrow \tan x = 1 \Rightarrow x = \pi / 4$.
- **Determine Relative Position of Curves:**
- • On $[0, \pi/4]$: $\cos x \geq \sin x$. Top is $\cos x$, bottom is $\sin x$.
- • On $[\pi/4, \pi]$: $\sin x \geq \cos x$. Top is $\sin x$, bottom is $\cos x$.
- **Split the Integral into Two Parts:**
- • $\text{Area}_1 = \int_0^{\pi/4} (\cos x - \sin x)\,dx = [\sin x + \cos x]_0^{(\pi/4)} = \sqrt{2} - 1$.
- • $\text{Area}_2 = \int_{\pi/4}^{\pi} (\sin x - \cos x)\,dx = [-\cos x - \sin x]_{(\pi/4)}^{\pi}$.
- • At $x = \pi$: $-\cos(\pi) - \sin(\pi) = -(-1) - 0 = 1$.
- • At $x = \pi/4$: $-\cos(\pi/4) - \sin(\pi/4) = -\sqrt{2}/2 - \sqrt{2}/2 = -\sqrt{2}$.
- • $\text{Area}_2 = 1 - (-\sqrt{2}) = 1 + \sqrt{2}$.
- **Sum the Two Areas:**
- • Total Area = $\text{Area}_1 + \text{Area}_2 = (\sqrt{2} - 1) + (1 + \sqrt{2}) = 2\sqrt{2}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Calculates Area_1 and doubles it: $2(\sqrt{2} - 1) = 2\sqrt{2} - 2$, failing to realize the two subregions are unequal.

✗ **(C)** Calculates only the difference of square root terms: $\sqrt{2}$.

✗ **(D)** Net signed integral evaluation error.

Question 46 Analysis

★★★★ MANDATORY

Correct Answer: (B) 4 / 3

STEP-BY-STEP SOLUTION

- **Find Points of Intersection with the y-axis:** $2y - y^2 = 0 \Rightarrow y(2 - y) = 0 \Rightarrow y = 0$ and $y = 2$.
- **Horizontal Slicing:** For $0 \le y \le 2$, $x = 2y - y^2 \ge 0$, so x_right = $2y - y^2$ and x_left = 0.
- **Set Up the Area Integral:** Area = $\int_0^2 (2y - y^2)\,dy$.
- **Evaluate the Definite Integral:**
- • $[y^2 - (1/3)y^3]_0^2$.
- • At y = 2: $(2)^2 - (1/3)(2)^3 = 4 - 8/3 = 12/3 - 8/3 = 4/3$.
- • At y = 0: 0.
- **Conclusion:** Area = 4/3.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to square y in the first term: 2 – 4/3 = 2/3.
✗ **(C)** Doubles the correct area: 8/3.
✗ **(D)** Approximates by a triangle with base 2 and height 1, yielding 1 or 2.

Question 47 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 2^(3/4)

STEP-BY-STEP SOLUTION

- **Total Area of the Region:** Total = $\int_0^2 y^3\,dy = [y^4/4]_0^2 = 16/4 = 4$.
- **Equal Area Condition:** The area from y = 0 to y = c must equal half of the total area.
- • Half Area = 4 / 2 = 2.
- **Set Up the Integral for c:**
- • $\int_0^c y^3\,dy = 2$.
- • $[y^4/4]_0^c = c^4/4 = 2$.
- **Solve for c:** $c^4 = 8 \Rightarrow c = 8^{(1/4)} = (2^3)^{(1/4)} = 2^{(3/4)}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Assumes midpoint of interval [0, 2] divides area equally: c = 1.
✗ **(B)** Solves $c^4 = 2$ instead of $c^4 = 8$, getting 2^(1/4).
✗ **(D)** Solves $c^2 = 2 \Rightarrow c = \sqrt{2}$, incorrectly using quadratic power.

Question 48 Analysis

★★★★ MANDATORY

Correct Answer: (D) 1 / 3

STEP-BY-STEP SOLUTION

- **Find Intersections in y:** $\sqrt{y} = y^2 \Rightarrow y = y^4 \Rightarrow y(y^3 - 1) = 0 \Rightarrow y = 0$ and $y = 1$.
- **Horizontal Slicing on [0, 1]:** For $0 < y < 1$, $\sqrt{y} > y^2$, so x_right = y^(1/2) and x_left = y^2.
- **Set Up Area Integral:** Area = $\int_0^1 (y^{(1/2)} - y^2)\,dy$.
- **Evaluate Antiderivative:**
- • $[(2/3)y^{(3/2)} - (1/3)y^3]_0^1$.
- • At y = 1: 2/3 – 1/3 = 1/3.
- • At y = 0: 0.
- **Conclusion:** Area = 1/3.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Arithmetic error in subtracting 2/3 – 1/2 = 1/6.
✗ **(B)** Omits the subtraction of the inner curve: reports 2/3.
✗ **(C)** Averages the powers or assumes triangle area (1/2)(1)(1) = 1/2.

Question 49 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 37 / 12

STEP-BY-STEP SOLUTION

- **Find Points of Intersection:**
- • $x^3 - 2x^2 = 2x - x^2 \Rightarrow x^3 - x^2 - 2x = 0$.
- • $x(x - 2)(x + 1) = 0 \Rightarrow x = -1, x = 0$, and $x = 2$.
- **Determine Relative Position of Curves:**
- • On $[-1, 0]$: Test $x = -0.5$. Top curve is $y = x^3 - 2x^2$, bottom is $y = 2x - x^2$.
- • On $[0, 2]$: Test $x = 1$. Top curve is $y = 2x - x^2$, bottom is $y = x^3 - 2x^2$.
- **Antiderivative of Difference $D(x) = x^3 - x^2 - 2x$:**
- • $\int (x^3 - x^2 - 2x)\,dx = (1/4)x^4 - (1/3)x^3 - x^2$.
- **Evaluate Area on $[-1, 0]$:**
- • $\text{Area}_1 = [(1/4)x^4 - (1/3)x^3 - x^2]_{-1}^{0} = 0 - [1/4 + 1/3 - 1] = -(7/12 - 1) = 5/12$.
- **Evaluate Area on $[0, 2]$:**
- • $\text{Area}_2 = -[(1/4)x^4 - (1/3)x^3 - x^2]_0^2 = -[(1/4)(16) - (1/3)(8) - 4] = -[4 - 8/3 - 4] = 8/3 = 32/12$.
- **Sum the Two Subregions:**
- • Total Area = $5/12 + 32/12 = 37/12$

Question 50 Analysis

★★★★ MANDATORY

Correct Answer: (B) 24.5

STEP-BY-STEP SOLUTION

- **Horizontal Slicing Area Formula:** Area = $\int_1^5 [f(y) - g(y)]\,dy$.
- **Compute the Width $w(y) = f(y) - g(y)$ at Each Grid Point:**
- • At $y = 1$: $w(1) = 7 - 1 = 6$.
- • At $y = 2$: $w(2) = 9 - 3 = 6$.
- • At $y = 3$: $w(3) = 11 - 4 = 7$.
- • At $y = 4$: $w(4) = 8 - 2 = 6$.
- • At $y = 5$: $w(5) = 6 - 1 = 5$.
- **Apply Trapezoidal Rule with $\Delta y = 1$:**
- • $\text{Area} \approx (\Delta y / 2) \cdot [w(1) + 2w(2) + 2w(3) + 2w(4) + w(5)]$.
- • $(1 / 2) \cdot [6 + 2(6) + 2(7) + 2(6) + 5]$.
- • $(1 / 2) \cdot [6 + 12 + 14 + 12 + 5] = (1 / 2) \cdot 49 = 24.5$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses a Left Riemann sum: $1 \cdot (6 + 6 + 7 + 6) = 25$ or calculation mistake yielding 27.0.

✗ **(C)** Uses a Right Riemann sum: $1 \cdot (6 + 7 + 6 + 5) = 24$ or misadds terms to 31.5.

Question 51 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 8

STEP-BY-STEP SOLUTION

- **Identify the Cross-Section Geometry:** Cross sections perpendicular to the x-axis are squares with side length $s(x) = \sqrt{x}$.
- **Area of One Cross Section:** $A(x) = [s(x)]^2 = (\sqrt{x})^2 = x$.
- **Set Up the Volume Integral:** Volume = $\int_0^4 A(x)\,dx = \int_0^4 x\,dx$.
- **Evaluate the Definite Integral:**
- • $[x^2 / 2]_0^4 = (4)^2 / 2 - 0 = 16 / 2 = 8$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Integrates using the volume of revolution formula $\pi \int (\sqrt{x})^2\,dx$ or treats as 1/3 base area.

✗ **(B)** Evaluates $[x^2]_0^4$ without dividing by 2: 16.

✗ **(D)** Averages the endpoint cross-sectional areas and multiplies by interval length: $((0 + 2)/2) \cdot 4 = 4$.

Question 52 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) $4\sqrt{3}/15$

STEP-BY-STEP SOLUTION

- **Find Boundaries in x:** $1 - x^2 = 0 \Rightarrow x = -1$ and $x = 1$.
- **Side Length of the Equilateral Triangle:** $s(x) = (1 - x^2) - 0 = 1 - x^2$.
- **Cross-Sectional Area Formula for Equilateral Triangle:**
- • Area $= (\sqrt{3}/4) \cdot s^2 = (\sqrt{3}/4) \cdot (1 - x^2)^2 = (\sqrt{3}/4) \cdot (1 - 2x^2 + x^4)$.
- **Set Up and Evaluate the Volume Integral:**
- • Volume $= \int_{-1}^{1} (\sqrt{3}/4)(1 - 2x^2 + x^4)\,dx = 2 \cdot (\sqrt{3}/4) \int_{0}^{1} (1 - 2x^2 + x^4)\,dx$.
- • $(\sqrt{3}/2)\left[x - (2/3)x^3 + (1/5)x^5\right]_0^1$.
- • At $x = 1$: $1 - 2/3 + 1/5 = 15/15 - 10/15 + 3/15 = 8/15$.
- • Multiply by $\sqrt{3}/2$: $(\sqrt{3}/2) \cdot (8/15) = 4\sqrt{3}/15$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses area formula $(\sqrt{3}/3) \cdot s^2$.

✗ **(B)** Forgets to simplify the factor $2/4 = 1/2$, retaining $2 \cdot (\sqrt{3}/4) \cdot (8/15) = 8\sqrt{3}/15$.

✗ **(C)** Arithmetic error in combining $1 - 2/3 + 1/5$, writing $6/15 = 2/5$.

Question 53 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $\pi^2/16$

STEP-BY-STEP SOLUTION

- **Find the Diameter and Radius of Semicircles:**
- • Diameter $d(x) = \sin x$.
- • Radius $r(x) = (1/2)\sin x$.
- **Area of Semicircle Cross Section:**
- • $A(x) = (1/2)\pi[r(x)]^2 = (1/2)\pi[(1/2)\sin x]^2 = (1/8)\pi\sin^2 x$.
- **Set Up the Volume Integral:**
- • Volume $= \int_{0}^{\pi} (\pi/8)\sin^2 x\,dx = (\pi/8)\int_{0}^{\pi} \sin^2 x\,dx$.
- **Evaluate the Definite Integral of $\sin^2 x$:**
- • Recall $\int_{0}^{\pi} \sin^2 x\,dx = \int_{0}^{\pi} (1 - \cos 2x)/2\,dx = [x/2 - \sin(2x)/4]_0^{\wedge}\pi = \pi/2$.
- **Compute the Final Volume:**
- • Volume $= (\pi/8) \cdot (\pi/2) = \pi^2/16$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses radius $r = \sin x$ instead of $r = (1/2)\sin x$, yielding $(\pi/2) \cdot (\pi/4)$ or $(\pi/4) \cdot (\pi/2) = \pi^2/8$.

✗ **(C)** Uses full circle area instead of semicircle area: $\pi^2/4$.

Question 54 Analysis

★★★★ MANDATORY

Correct Answer: (B) $(e^2 - 1)/2$

STEP-BY-STEP SOLUTION

- **Side Length of the Square:** $s(x) = e^x$.
- **Area of Square Cross Section:** $A(x) = [s(x)]^2 = (e^x)^2 = e^{2x}$.
- **Set Up Volume Integral:** Volume $= \int_{0}^{1} e^{2x}\,dx$.
- **Evaluate Antiderivative:**
- • $\left[(1/2)e^{2x}\right]_0^1 = (1/2)e^2 - (1/2)e^0 = (e^2 - 1)/2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the factor of 1/2 from chain rule / u-substitution on e^{2x}: $e^2 - 1$.

✗ **(C)** Divides by 4 instead of 2.

✗ **(D)** Neglects to subtract lower limit evaluation at $x = 0$ (where $e^0 = 1$).

Question 55 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 32

STEP-BY-STEP SOLUTION

- **Identify Direction of Slicing:** Slicing is perpendicular to the y-axis, so integrate with respect to y from $y = 0$ to $y = 4$.
- **Side Length of Square in terms of y:**
- • The curve is $x = \pm\sqrt{y}$.
- • Side length $s(y) = (\sqrt{y}) - (-\sqrt{y}) = 2\sqrt{y}$.
- **Area of Square Cross Section:**
- • $A(y) = [s(y)]^2 = (2\sqrt{y})^2 = 4y$.
- **Set Up and Evaluate the Volume Integral:**
- • Volume = $\int_0^4 4y\,dy = [\,2y^2\,]_0^4 = 2(4)^2 - 0 = 2(16) = 32$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses only one half of the base: $s(y) = \sqrt{y} \Rightarrow A(y) = y \Rightarrow [y^2/2]_0^4 = 8$, or miscalculates to 16.

✗ **(B)** Uses $s(y) = 4\sqrt{y}$: $A(y) = 16y \Rightarrow 8(16) = 128$, or forgets 1/2 in antiderivative: $4(16) = 64$.

✗ **(D)** Integrates with respect to x instead of y: $\int_{-2}^{2} (4 - x^2)^2\,dx = 256/15$.

Question 56 Analysis

★★★★ MANDATORY

Correct Answer: (D) 2 / 3

STEP-BY-STEP SOLUTION

- **Find Base Boundaries:** The triangle extends along the x-axis from $x = 0$ to $x = 2$.
- **Hypotenuse of Cross Section:** $h(x) = 2 - x$.
- **Area of Isosceles Right Triangle with Hypotenuse h:**
- • If hypotenuse is h, legs are $h / \sqrt{2}$.
- • Area $A(x) = (1/2) \cdot \text{leg} \cdot \text{leg} = (1/2) \cdot (h / \sqrt{2})^2 = (1/4)\,h^2$.
- • $A(x) = (1/4)(2 - x)^2$.
- **Set Up and Evaluate the Volume Integral:**
- • Volume = $\int_0^2 (1/4)(2 - x)^2\,dx$.
- • Let $u = 2 - x \Rightarrow du = -dx$.
- • $[\,-(1/12)\,(2 - x)^3\,]_0^2 = 0 - [\,-(1/12)(2)^3\,] = (1/12)(8) = 8 / 12 = 2 / 3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Treats cross section as a square: 8/3.

✗ **(B)** Treats the base as the leg of the triangle instead of the hypotenuse: $A(x) = (1/2)(2 - x)^2 \Rightarrow$ Volume = 4/3.

✗ **(C)** Halves the correct volume: 1/3.

Question 57 Analysis

★★★ IMPORTANT

Correct Answer: (A) 12

STEP-BY-STEP SOLUTION

- **Dimensions of the Rectangular Cross Section:**
- • Base = $b(x) = 2 \sin x$.
- • Height = $h = 3$.
- **Area of Cross Section:** $A(x) = b(x) \cdot h = (2 \sin x)(3) = 6 \sin x$.
- **Set Up the Volume Integral:** Volume = $\int_0^{\pi} 6 \sin x\,dx$.
- **Evaluate Antiderivative:**
- • $[\,-6 \cos x\,]_0\text{^}\pi = (-6 \cos \pi) - (-6 \cos 0) = -6(-1) - (-6)(1) = 6 + 6 = 12$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets that $-\cos \pi - (-\cos 0) = 1 - (-1) = 2$, multiplying by 1 instead: 6.

✗ **(C)** Calculation mistake evaluating $6(3) = 18$.

✗ **(D)** Includes a factor of π from revolving, confusing cross sections with solids of revolution.

Question 58 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 144

STEP-BY-STEP SOLUTION

- **Determine Side Length of the Square:**
- • For a given $x \in [-3, 3]$, y ranges from $-\sqrt{(9 - x^2)}$ to $+\sqrt{(9 - x^2)}$.
- • Side length $s(x) = 2\sqrt{(9 - x^2)}$.
- **Cross-Sectional Area:**
- • $A(x) = [s(x)]^2 = [2\sqrt{(9 - x^2)}]^2 = 4(9 - x^2)$.
- **Set Up the Volume Integral:**
- • Volume = $\int_{-3}^{3} 4(9 - x^2)\,dx = 2\int_{0}^{3} 4(9 - x^2)\,dx = 8\int_{0}^{3} (9 - x^2)\,dx$.
- **Evaluate the Definite Integral:**
- • $8[9x - (1/3)x^3]_0^3$.
- • At $x = 3$: $9(3) - (1/3)(27) = 27 - 9 = 18$.
- • Final Volume = $8 \cdot 18 = 144$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses side length $s(x) = \sqrt{(9 - x^2)}$, yielding $A(x) = 9 - x^2 \Rightarrow$ Volume = 36 or 72.

✗ **(C)** Confuses cross-sectional squares with a sphere of radius 3: $(4/3)\pi(27) = 36\pi$.

Question 59 Analysis

★★★★ MANDATORY

Correct Answer: (C) 65

STEP-BY-STEP SOLUTION

- **Volume as an Integral of Cross-Sectional Area:** Volume = $\int_{0}^{8} A(x)\,dx$.
- **Apply Trapezoidal Rule with Uniform Width $\Delta x = 2$:**
- • Volume $\approx (\Delta x / 2) \cdot [A(0) + 2A(2) + 2A(4) + 2A(6) + A(8)]$.
- • $(2 / 2) \cdot [3 + 2(7) + 2(12) + 2(10) + 4]$.
- • $1 \cdot [3 + 14 + 24 + 20 + 4]$.
- • $3 + 14 = 17$; $17 + 24 = 41$; $41 + 20 = 61$; $61 + 4 = 65$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses a Left Riemann sum: $2(3 + 7 + 12 + 10) = 64$, or arithmetic mistake giving 72.

✗ **(B)** Forgets to divide by 2 in the trapezoidal rule factor $(\Delta x / 2)$, multiplying by 2: 130.

✗ **(D)** Uses a Right Riemann sum: $2(7 + 12 + 10 + 4) = 66$, or miscalculation giving 58.

Question 60 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) (e – 2) / 2

STEP-BY-STEP SOLUTION

- **Identify Boundaries in x:** $\ln x = 0 \Rightarrow x = 1$. The region extends from $x = 1$ to $x = e$.
- **Leg Length of the Cross Section:** Leg = $y = \ln x$.
- **Area of Isosceles Right Triangle with Leg in Base:** $A(x) = (1/2) \cdot \text{leg}^2 = (1/2)(\ln x)^2$.
- **Set Up the Volume Integral:** Volume = $(1/2)\int_{1}^{e} (\ln x)^2\,dx$.
- **Evaluate the Integral using Integration by Parts:**
- • Recall $\int (\ln x)^2\,dx = x(\ln x)^2 - 2x \ln x + 2x$.
- • At $x = e$: $e(1)^2 - 2e(1) + 2e = e - 2e + 2e = e$.
- • At $x = 1$: $1(0) - 2(1)(0) + 2(1) = 2$.
- • Value of definite integral = $e - 2$.
- **Multiply by 1/2:** Volume = $(1/2)(e - 2) = (e - 2) / 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Neglects the lower limit evaluation at $x = 1$, giving $e/2$.

✗ **(B)** Forgets the 1/2 factor in the area of a right triangle: $e - 2$.

✗ **(C)** Arithmetic error evaluating antiderivative at lower limit: $(e - 1)/2$.

Question 61 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 8π

STEP-BY-STEP SOLUTION

- **Identify Radius of Revolution:** When revolving about the x-axis, the cross-sectional slice is a circular disk with radius $R(x) = \sqrt{x}$.
- **Area of Circular Disk:** $A(x) = \pi [R(x)]^2 = \pi (\sqrt{x})^2 = \pi x$.
- **Set Up the Volume Integral:** Volume = $\int_0^4 \pi x \, dx = \pi [x^2 / 2]_0^4$.
- **Evaluate the Definite Integral:**
- • $\pi [(4)^2 / 2 - 0] = \pi (16 / 2) = 8\pi$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets to divide by 2 in the antiderivative of x: 16π.
✗ **(C)** Halves the correct volume: 4π.
✗ **(D)** Uses a formula for a cone or sphere: $32\pi / 3$.

Question 62 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 8π

STEP-BY-STEP SOLUTION

- **Identify Axis of Revolution and Radius:** Revolving about the y-axis requires integrating with respect to y.
- • Solve $y = x^2$ for x: $x = \sqrt{y}$ (since $x \geq 0$).
- • Radius $R(y) = \sqrt{y}$.
- **Area of Circular Disk:** $A(y) = \pi [R(y)]^2 = \pi (\sqrt{y})^2 = \pi y$.
- **Limits of Integration:** y ranges from $y = 0$ to $y = 4$.
- **Evaluate the Volume Integral:**
- • Volume = $\pi \int_0^4 y \, dy = \pi [y^2 / 2]_0^4 = \pi (16 / 2) = 8\pi$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the 1/2 factor in the antiderivative: 16π.
✗ **(C)** Integrates with respect to x around the y-axis using an incorrect formula: $32\pi / 3$.
✗ **(D)** Revolves about the x-axis instead: $\pi \int_0^2 (4 - x^2)^2 \, dx$ or $\pi \int_0^2 (x^2)^2 \, dx = 32\pi / 5$.

Question 63 Analysis

★★★★ MANDATORY

Correct Answer: (C) $\pi(e^2 - 1) / 2$

STEP-BY-STEP SOLUTION

- **Radius of Revolution:** $R(x) = e^x$.
- **Disk Area:** $A(x) = \pi [R(x)]^2 = \pi (e^x)^2 = \pi e^{2x}$.
- **Set Up the Volume Integral:** Volume = $\pi \int_0^1 e^{2x} \, dx$.
- **Evaluate the Definite Integral:**
- • $\pi [(1/2) e^{2x}]_0^1 = (\pi / 2) [e^2 - e^0] = \pi(e^2 - 1) / 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Neglects the lower limit evaluation at $x = 0$ ($e^0 = 1$): $\pi e^2 / 2$.
✗ **(B)** Forgets the 1/2 factor from the exponent $2x$: $\pi(e^2 - 1)$.
✗ **(D)** Multiplies by 2 instead of dividing by 2: $2\pi(e^2 - 1)$.

Question 64 Analysis

★★★★ MANDATORY

Correct Answer: (D) $8\pi / 3$

STEP-BY-STEP SOLUTION

- **Integration Boundaries:** The line $y = 2 - x$ intersects the x-axis at $x = 2$ and y-axis at $(0, 2)$. Thus $x \in [0, 2]$.
- **Radius of Revolution:** $R(x) = 2 - x$.
- **Disk Method Integral:** Volume $= \pi \int_0^2 (2 - x)^2 \, dx$.
- **Evaluate Using Substitution:**
- • Let $u = 2 - x \Rightarrow du = -dx$.
- • $\pi [-(1/3)(2 - x)^3]_0^2 = \pi [0 - (-(1/3)(2)^3)] = \pi (8 / 3) = 8\pi / 3$.
- **Geometric Check (Right Circular Cone):**
- • Radius $r = 2$, Height $h = 2$.
- • $V = (1/3) \pi r^2 h = (1/3) \pi (2)^2 (2) = 8\pi / 3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Doubles the correct volume: $16\pi / 3$.

✗ **(B)** Halves the correct volume or uses height 1: $4\pi / 3$.

✗ **(C)** Calculates the cylinder volume $\pi r^2 h = 8\pi$ without the 1/3 cone factor.

Question 65 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $\pi^2 / 4$

STEP-BY-STEP SOLUTION

- **Radius of Revolution:** $R(x) = \cos x$.
- **Disk Area:** $A(x) = \pi [\cos x]^2 = \pi \cos^2 x$.
- **Set Up the Volume Integral:** Volume $= \pi \int_0^{\pi/2} \cos^2 x \, dx$.
- **Apply Half-Angle Identity:** $\cos^2 x = (1 + \cos 2x) / 2$.
- **Evaluate the Definite Integral:**
- • $\pi \int_0^{\pi/2} (1/2 + (1/2) \cos 2x) \, dx = \pi [x / 2 + (1/4) \sin 2x]_0^{(\pi/2)}$.
- • At $x = \pi/2$: $(\pi/2)/2 + (1/4) \sin(\pi) = \pi/4 + 0 = \pi/4$.
- • At $x = 0$: $0 + 0 = 0$.
- • Volume $= \pi \cdot (\pi / 4) = \pi^2 / 4$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Integrates cos x without squaring: $\pi \int_0^{\pi/2} \cos x \, dx = \pi(1) = \pi$ or miscalculates to $\pi/2$.

✗ **(C)** Omits the 1/2 factor in the half-angle identity: $\pi^2 / 2$.

✗ **(D)** Forgets the exterior factor of π from the disk method: $\pi / 4$.

Question 66 Analysis

★★★★ MANDATORY

Correct Answer: (B) $32\pi / 3$

STEP-BY-STEP SOLUTION

- **Find Boundaries in x:** $\sqrt{(4 - x^2)} = 0 \Rightarrow x = -2$ and $x = 2$.
- **Disk Radius:** $R(x) = \sqrt{(4 - x^2)}$.
- **Disk Area:** $A(x) = \pi [R(x)]^2 = \pi (4 - x^2)$.
- **Set Up and Evaluate Volume Integral:**
- • Volume $= \pi \int_{-2}^{2} (4 - x^2) \, dx = 2\pi \int_0^2 (4 - x^2) \, dx$.
- • $2\pi [4x - (1/3) x^3]_0^2 = 2\pi [4(2) - (1/3)(8)] = 2\pi [8 - 8/3] = 2\pi (16 / 3) = 32\pi / 3$.
- **Geometric Check:** Sphere of radius $r = 2$ has volume $V = (4/3) \pi r^3 = (4/3) \pi (8) = 32\pi / 3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Integrates only over the right hemisphere $[0, 2]$, forgetting to double: $16\pi / 3$.

✗ **(C)** Forgets the 1/3 factor in the antiderivative of x^2: $2\pi(8 - 0) = 16\pi$.

✗ **(D)** Uses diameter 4 instead of radius 2 in sphere volume formula: $64\pi / 3$.

Question 67 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 512π / 15

STEP-BY-STEP SOLUTION

- **Identify Boundaries in x:** $4 - x^2 = 0 \Rightarrow x = -2$ and $x = 2$.
- **Radius of Revolution:** $R(x) = 4 - x^2$.
- **Set Up the Volume Integral:**
- • Volume $= \pi \int_{-2}^{2} (4 - x^2)^2 \, dx = 2\pi \int_{0}^{2} (16 - 8x^2 + x^4) \, dx$.
- **Evaluate Antiderivative:**
- • $2\pi \left[16x - (8/3)x^3 + (1/5)x^5 \right]_0^2$.
- • At $x = 2$: $16(2) - (8/3)(8) + (1/5)(32) = 32 - 64/3 + 32/5$.
- • Factor out 32: $32 \cdot [1 - 2/3 + 1/5] = 32 \cdot [15/15 - 10/15 + 3/15] = 32 \cdot (8/15) = 256/15$.
- • Multiply by 2π: $2\pi \cdot (256/15) = 512\pi/15$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Arithmetic error combining fractions: $128\pi/5$.
✗ **(B)** Integrates only over $[0, 2]$ without doubling for symmetry: $256\pi/15$.
✗ **(D)** Revolves a triangle approximation: $64\pi/3$.

Question 68 Analysis

★★★★ MANDATORY

Correct Answer: (D) 2π / 3

STEP-BY-STEP SOLUTION

- **Axis of Revolution:** y-axis. Integrate with respect to y from $y = 1$ to $y = 3$.
- **Disk Radius:** $R(y) = 1/y$.
- **Disk Area:** $A(y) = \pi [R(y)]^2 = \pi (1/y)^2 = \pi y^{-2}$.
- **Set Up and Evaluate the Volume Integral:**
- • Volume $= \pi \int_{1}^{3} y^{-2} \, dy = \pi \left[-1/y \right]_1^3$.
- • At $y = 3$: $-1/3$.
- • At $y = 1$: $-1/1 = -1$.
- • Difference: $-1/3 - (-1) = -1/3 + 1 = 2/3$.
- • Multiply by π: Volume $= 2\pi/3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Integrates $1/y$ without squaring, using natural logarithm: $\pi \ln 3$.
✗ **(B)** Arithmetic error in subtracting $-1/3 - (-1)$, writing $1/3$: $\pi/3$.
✗ **(C)** Doubles the correct volume: $4\pi/3$.

Question 69 Analysis

★★★ IMPORTANT

Correct Answer: (A) 9π

STEP-BY-STEP SOLUTION

- **Express Boundary in terms of y:** $y = 3 - x \Rightarrow x = 3 - y$.
- **Limits of Integration in y:** The region extends from $y = 0$ to $y = 3$.
- **Disk Radius:** $R(y) = 3 - y$.
- **Volume Integral:** Volume $= \pi \int_{0}^{3} (3 - y)^2 \, dy$.
- **Evaluate Antiderivative:**
- • $\pi \left[-(1/3)(3 - y)^3 \right]_0^3 = \pi \left[0 - (-(1/3)(3)^3) \right] = \pi (27/3) = 9\pi$.
- **Geometric Check:** Cone with base radius $r = 3$ and height $h = 3$ has volume $V = (1/3)\pi (3)^2 (3) = 9\pi$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Calculates the cylinder volume $\pi r^2 h = 27\pi$ forgetting the 1/3 cone factor.
✗ **(C)** Multiplies by 2 instead of 1/3: 18π.
✗ **(D)** Divides 9π by 2 instead of computing the full revolution: $9\pi/2$.

Question 70 Analysis

★★★★ MANDATORY

Correct Answer: (B) 55π

STEP-BY-STEP SOLUTION

- **Disk Method Volume Formula:** Volume = $\pi \int_0^6 [R(x)]^2\,dx$.
- **Compute $[R(x)]^2$ at Each Given x:**
- • At x = 0: $[R(0)]^2 = (1)^2 = 1$.
- • At x = 2: $[R(2)]^2 = (3)^2 = 9$.
- • At x = 4: $[R(4)]^2 = (4)^2 = 16$.
- • At x = 6: $[R(6)]^2 = (2)^2 = 4$.
- **Apply Trapezoidal Rule with Δx = 2:**
- • $\int_0^6 [R(x)]^2\,dx \approx (\Delta x / 2) \cdot [1 + 2(9) + 2(16) + 4]$.
- • $(2 / 2) \cdot [1 + 18 + 32 + 4] = 1 \cdot 55 = 55$.
- **Multiply by π:** Volume ≈ 55π.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to divide by 2 in the trapezoidal rule factor (Δx / 2), multiplying by 2: 110π.
✗ **(C)** Uses a Left Riemann sum: 2·(1 + 9 + 16)π = 52π or calculation error giving 60π.
✗ **(D)** Approximates by squaring the sum of radii rather than summing squares of radii.

Question 71 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 2π / 15

STEP-BY-STEP SOLUTION

- **Find Points of Intersection:** $x = x^2 \Rightarrow x(1 - x) = 0 \Rightarrow x = 0$ and $x = 1$.
- **Identify Outer and Inner Radii on [0, 1]:**
- • For $0 < x < 1$, $x > x^2$.
- • Outer radius: $R(x) = x$.
- • Inner radius: $r(x) = x^2$.
- **Washer Method Formula:** Volume = $\pi \int_0^1 [(R(x))^2 - (r(x))^2]\,dx$.
- **Set Up and Evaluate the Integral:**
- • Volume = $\pi \int_0^1 [x^2 - (x^2)^2]\,dx = \pi \int_0^1 (x^2 - x^4)\,dx$.
- • $\pi [(1/3)x^3 - (1/5)x^5]_0^1 = \pi [1/3 - 1/5] = \pi (5/15 - 3/15) = 2\pi / 15$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Doubles the correct volume: 4π / 15.
✗ **(B)** Halves the correct volume: π / 15.
✗ **(D)** Revolves under line only without subtracting inner curve: $\pi \int_0^1 x^2\,dx = \pi / 3$.

Question 72 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) π / 6

STEP-BY-STEP SOLUTION

- **Express Curves in terms of y:**
- • $y = x^2 \Rightarrow x = \sqrt{y}$ (outer boundary).
- • $y = x \Rightarrow x = y$ (inner boundary).
- **Limits of Integration in y:** y ranges from y = 0 to y = 1.
- **Outer and Inner Radii:**
- • Outer radius: $R(y) = \sqrt{y}$.
- • Inner radius: $r(y) = y$.
- **Set Up the Washer Integral:**
- • Volume = $\pi \int_0^1 [(\sqrt{y})^2 - y^2]\,dy = \pi \int_0^1 (y - y^2)\,dy$.
- **Evaluate the Definite Integral:**
- • $\pi [(1/2)y^2 - (1/3)y^3]_0^1 = \pi [1/2 - 1/3] = \pi (3/6 - 2/6) = \pi / 6$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Halves the correct volume: π / 12.
✗ **(B)** Forgets to subtract inner radius: $\pi \int_0^1 y\,dy = \pi / 2$ or arithmetic mistake giving π / 3.
✗ **(C)** Confuses revolution about y-axis with revolution about x-axis (which yields 2π / 15).

Question 73 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 136π / 15

STEP-BY-STEP SOLUTION

- **Find Points of Intersection:** $2x = x^2 \Rightarrow x = 0$ and $x = 2$.
- **Identify Radii Relative to the Line y = 5:**
- • Since $y = 5$ lies above both curves for all $x \in [0, 2]$:
- • Outer radius (distance to farther curve $y = x^2$): $R(x) = 5 - x^2$.
- • Inner radius (distance to closer curve $y = 2x$): $r(x) = 5 - 2x$.
- **Compute the Difference of Squares:**
- • $[R(x)]^2 - [r(x)]^2 = (5 - x^2)^2 - (5 - 2x)^2$.
- • $(25 - 10x^2 + x^4) - (25 - 20x + 4x^2) = x^4 - 14x^2 + 20x$.
- **Evaluate the Volume Integral:**
- • Volume $= \pi \int_0^2 (x^4 - 14x^2 + 20x)\,dx = \pi \left[(1/5)x^5 - (14/3)x^3 + 10x^2 \right]_0^2$.
- • At $x = 2$: $(1/5)(32) - (14/3)(8) + 10(4) = 32/5 - 112/3 + 40$.
- • Combine terms: $32/5 + (-112/3 + 120/3) = 32/5 + 8/3 = (96 + 40) / 15 = 136 / 15$.
- • Volume $= 136\pi / 15$.

WHY OTHER OPTIONS ARE INCORRECT

Question 74 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 832π / 15

STEP-BY-STEP SOLUTION

- **Find Boundaries in x:** $4 - x^2 = 0 \Rightarrow x = -2$ and $x = 2$.
- **Identify Radii Relative to y = −1:**
- • The axis $y = -1$ lies below the region.
- • Outer radius: $R(x) = (4 - x^2) - (-1) = 5 - x^2$.
- • Inner radius: $r(x) = 0 - (-1) = 1$.
- **Compute $[R(x)]^2 - [r(x)]^2$:**
- • $(5 - x^2)^2 - (1)^2 = (25 - 10x^2 + x^4) - 1 = 24 - 10x^2 + x^4$.
- **Set Up Integral and Use Symmetry:**
- • Volume $= \pi \int_{-2}^{2} (24 - 10x^2 + x^4)\,dx = 2\pi \int_0^2 (24 - 10x^2 + x^4)\,dx$.
- • $2\pi \left[24x - (10/3)x^3 + (1/5)x^5 \right]_0^2$.
- • At $x = 2$: $24(2) - (10/3)(8) + (1/5)(32) = 48 - 80/3 + 32/5$.
- • Common denominator 15: $720/15 - 400/15 + 96/15 = 416 / 15$.
- • Multiply by 2π: $2\pi \cdot (416 / 15) = 832\pi / 15$.

WHY OTHER OPTIONS ARE INCORRECT

Question 75 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 136π / 15

STEP-BY-STEP SOLUTION

- **Find Boundaries in y:** $y^2 = 2y \Rightarrow y = 0$ and $y = 2$.
- **Identify Radii Relative to x = 5:**
- • Vertical axis $x = 5$ lies to the right of the region for all $y \in [0, 2]$.
- • Outer radius (distance to farther curve $x = y^2$): $R(y) = 5 - y^2$.
- • Inner radius (distance to closer curve $x = 2y$): $r(y) = 5 - 2y$.
- **Compute Difference of Squares:**
- • $[R(y)]^2 - [r(y)]^2 = (5 - y^2)^2 - (5 - 2y)^2 = y^4 - 14y^2 + 20y$.
- **Evaluate the Volume Integral:**
- • Volume $= \pi \int_0^2 (y^4 - 14y^2 + 20y)\,dy = \pi \left[(1/5)y^5 - (14/3)y^3 + 10y^2 \right]_0^2$.
- • Evaluates to exactly $136\pi / 15$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Arithmetic error in evaluating polynomial at $y = 2$: $112\pi / 15$.

✗ **(B)** Halves the correct volume: $68\pi / 15$.

✗ **(D)** Reverses inner and outer radii.

Question 76 Analysis

★★★★ MANDATORY

Correct Answer: (D) $8\pi / 3$

STEP-BY-STEP SOLUTION

- **Find Points of Intersection:** $\sqrt{x} = x/2 \Rightarrow x = x^2/4 \Rightarrow x(x-4) = 0 \Rightarrow x = 0$ and $x = 4$.
- **Outer and Inner Radii:**
- • On $(0, 4)$, $\sqrt{x} > x/2$.
- • Outer radius: $R(x) = \sqrt{x}$.
- • Inner radius: $r(x) = x/2$.
- **Set Up the Washer Integral:**
- • Volume $= \pi \int_0^4 [(\sqrt{x})^2 - (x/2)^2]\,dx = \pi \int_0^4 (x - x^2/4)\,dx$.
- **Evaluate Antiderivative:**
- • $\pi [(1/2)x^2 - (1/12)x^3]_0^4$.
- • At $x = 4$: $(1/2)(16) - (1/12)(64) = 8 - 16/3 = 24/3 - 16/3 = 8/3$.
- • Volume $= 8\pi/3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to subtract inner radius: $\pi \int_0^4 x\,dx = 8\pi$.

✗ **(B)** Halves the correct volume: $4\pi/3$.

Question 77 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $\pi(e^2 + 8e - 19)/2$

STEP-BY-STEP SOLUTION

- **Determine Integration Limits:** $e^x = 1 \Rightarrow x = 0$. The region extends from $x = 0$ to $x = 1$.
- **Identify Radii Relative to y = −2:**
- • Axis $y = -2$ lies below the region.
- • Outer radius: $R(x) = e^x - (-2) = e^x + 2$.
- • Inner radius: $r(x) = 1 - (-2) = 3$.
- **Compute $[R(x)]^2 - [r(x)]^2$:**
- • $(e^x + 2)^2 - 3^2 = (e^{2x} + 4e^x + 4) - 9 = e^{2x} + 4e^x - 5$.
- **Evaluate the Volume Integral:**
- • Volume $= \pi \int_0^1 (e^{2x} + 4e^x - 5)\,dx = \pi [(1/2)e^{2x} + 4e^x - 5x]_0^1$.
- • At $x = 1$: $(1/2)e^2 + 4e - 5$.
- • At $x = 0$: $(1/2)(1) + 4(1) - 0 = 1/2 + 4 = 9/2$.
- • Difference: $((1/2)e^2 + 4e - 5) - 9/2 = (1/2)e^2 + 4e - 19/2 = (e^2 + 8e - 19)/2$.
- • Volume $= \pi(e^2 + 8e - 19)/2$.

WHY OTHER OPTIONS ARE INCORRECT

Question 78 Analysis

★★★★ MANDATORY

Correct Answer: (B) $\pi(1/2 + 2\ln 2)$

STEP-BY-STEP SOLUTION

- **Identify Radii Relative to y = −1:**
- • Outer radius: $R(x) = (1/x) - (-1) = 1/x + 1$.
- • Inner radius: $r(x) = 0 - (-1) = 1$.
- **Compute Difference of Squares:**
- • $[R(x)]^2 - [r(x)]^2 = (1/x + 1)^2 - 1^2 = (1/x^2 + 2/x + 1) - 1 = 1/x^2 + 2/x$.
- **Evaluate the Volume Integral:**
- • Volume $= \pi \int_1^2 (x^{-2} + 2/x)\,dx = \pi [-1/x + 2\ln x]_1^2$.
- • At $x = 2$: $-1/2 + 2\ln 2$.
- • At $x = 1$: $-1 + 0 = -1$.
- • Difference: $(-1/2 + 2\ln 2) - (-1) = 1/2 + 2\ln 2$.
- • Volume $= \pi(1/2 + 2\ln 2)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the factor of 2 in the cross-term $2/x$: $\pi(1/2 + \ln 2)$.

✗ **(C)** Arithmetic error evaluating $-1/2 - (-1)$: $3/4$.

✗ **(D)** Evaluates upper limit as 0 instead of $-1/2$.

Question 79 Analysis

★★★ IMPORTANT

Correct Answer: (C) $\pi \int_0^{\pi} [(\sin x + 2)^2 - 4]\,dx$

STEP-BY-STEP SOLUTION

- **Identify the Axis of Revolution:** $y = -2$.
- **Determine Outer and Inner Radii:**
- • For $x \in [0, \pi]$, $y = \sin x$ lies above $y = 0$.
- • Outer radius $R(x) = \sin x - (-2) = \sin x + 2$.
- • Inner radius $r(x) = 0 - (-2) = 2$.
- **Washer Method Formulation:**
- • Volume $= \pi \int_0^{\pi} [(R(x))^2 - (r(x))^2]\,dx$.
- • Volume $= \pi \int_0^{\pi} [(\sin x + 2)^2 - 2^2]\,dx = \pi \int_0^{\pi} [(\sin x + 2)^2 - 4]\,dx$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Squares the individual terms rather than finding the radius: $(\sin x)^2 - 4$.
✗ **(B)** Forgets to subtract the inner radius squared (hole in the washer): $\pi \int (\sin x + 2)^2\,dx$.
✗ **(D)** Uses an incorrect sign in the radius: $\sin x - 2$.

Question 80 Analysis

★★★★ MANDATORY

Correct Answer: (D) 126.5π

STEP-BY-STEP SOLUTION

- **Washer Area at Each Point:** $A(x) = \pi [(R(x))^2 - (r(x))^2]$.
- **Compute $[R(x)]^2 - [r(x)]^2$ at Each Given x:**
- • At $x = 1$: $5^2 - 2^2 = 25 - 4 = 21$.
- • At $x = 2$: $6^2 - 2^2 = 36 - 4 = 32$.
- • At $x = 3$: $7^2 - 3^2 = 49 - 9 = 40$.
- • At $x = 4$: $6^2 - 2^2 = 36 - 4 = 32$.
- • At $x = 5$: $5^2 - 1^2 = 25 - 1 = 24$.
- **Apply Trapezoidal Rule with Δx = 1:**
- • Integral $\approx (\Delta x / 2) \cdot [A_1(1) + 2A_1(2) + 2A_1(3) + 2A_1(4) + A_1(5)]$.
- • $(1/2) \cdot [21 + 2(32) + 2(40) + 2(32) + 24]$.
- • $(1/2) \cdot [21 + 64 + 80 + 64 + 24] = (1/2) \cdot 253 = 126.5$.
- **Multiply by π:** Volume $\approx 126.5\pi$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses a Right Riemann sum: $1(32 + 40 + 32 + 24)\pi = 128\pi$ or arithmetic mistake

Question 81 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 14 / 3

STEP-BY-STEP SOLUTION

- **Recall Arc Length Formula:** $L = \int_a^b \sqrt{1 + [f'(x)]^2}\,dx$.
- **Compute the Derivative:**
- • $f(x) = (2/3)\,x^{(3/2)} \Rightarrow f'(x) = (2/3) \cdot (3/2)\,x^{(1/2)} = \sqrt{x}$.
- **Compute the Arc Length Integrand:**
- • $1 + [f'(x)]^2 = 1 + (\sqrt{x})^2 = 1 + x$.
- • $\sqrt{1 + [f'(x)]^2} = \sqrt{(1 + x)} = (1 + x)^{(1/2)}$.
- **Evaluate the Definite Integral from x = 0 to x = 3:**
- • $L = \int_0^3 (1 + x)^{(1/2)}\,dx = [(2/3)(1 + x)^{(3/2)}]_0^3$.
- • At $x = 3$: $(2/3)(1 + 3)^{(3/2)} = (2/3)(4)^{(3/2)} = (2/3)(8) = 16/3$.
- • At $x = 0$: $(2/3)(1 + 0)^{(3/2)} = (2/3)(1) = 2/3$.
- • $L = 16/3 - 2/3 = 14/3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Neglects to subtract lower limit evaluation at $x = 0$ $(2/3)$: $16/3$.
✗ **(C)** Halves the correct arc length: $7/3$.
✗ **(D)** Forgets the $2/3$ coefficient when integrating: $8 - 1 = 7$, or miscalculates to 8.

Question 82 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 17 / 12

STEP-BY-STEP SOLUTION

- **Compute Derivative y':**
- • $y = (1/6)x^3 + (1/2)x^{-1}$.
- • $y' = (1/2)x^2 - (1/2)x^{-2} = (x^2/2) - (1/(2x^2))$.
- **Algebraic Simplification of $1 + (y')^2$:**
- • $(y')^2 = (x^4/4) - 1/2 + (1/(4x^4))$.
- • $1 + (y')^2 = (x^4/4) + 1/2 + (1/(4x^4)) = [(x^2/2) + (1/(2x^2))]^2$.
- **Take Square Root:**
- • $\sqrt{(1 + (y')^2)} = (x^2/2) + (1/(2x^2))$ on $[1, 2]$.
- **Set Up and Evaluate the Integral:**
- • $L = \int_1^2 [(1/2)x^2 + (1/2)x^{-2}]\,dx = [(1/6)x^3 - (1/2)x^{-1}]_1^2$.
- • At $x = 2$: $(8/6) - (1/4) = 4/3 - 1/4 = 16/12 - 3/12 = 13/12$.
- • At $x = 1$: $(1/6) - (1/2) = 1/6 - 3/6 = -2/6 = -1/3 = -4/12$.
- • $L = 13/12 - (-4/12) = 17/12$.

WHY OTHER OPTIONS ARE INCORRECT

Question 83 Analysis

★★★★ MANDATORY

Correct Answer: (C) $\int_0^{\pi} \sqrt{(1 + 4\cos^2(2x))}\,dx$

STEP-BY-STEP SOLUTION

- **Find Derivative using Chain Rule:** $y = \sin(2x) \Rightarrow y' = 2\cos(2x)$.
- **Compute $[y']^2$:** $[y']^2 = [2\cos(2x)]^2 = 4\cos^2(2x)$.
- **Arc Length Formula:** $L = \int_a^b \sqrt{(1 + [y']^2)}\,dx$.
- • With $a = 0$ and $b = \pi$: $L = \int_0^{\pi} \sqrt{(1 + 4\cos^2(2x))}\,dx$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to square the derivative: $1 + 2\cos(2x)$.

✗ **(B)** Forgets chain rule factor 2 when differentiating $\sin(2x)$: $\cos^2(2x)$.

✗ **(D)** Differentiates sin to sin instead of cos.

Question 84 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) 4 / 3

STEP-BY-STEP SOLUTION

- **Arc Length in terms of y:** $L = \int_c^d \sqrt{(1 + [dx/dy]^2)}\,dy$.
- **Differentiate x with respect to y:**
- • $dx/dy = (1/3) \cdot (3/2)(y^2 + 2)^{(1/2)} \cdot (2y) = y\sqrt{(y^2 + 2)}$.
- **Compute $1 + [dx/dy]^2$:**
- • $[dx/dy]^2 = y^2(y^2 + 2) = y^4 + 2y^2$.
- • $1 + [dx/dy]^2 = y^4 + 2y^2 + 1 = (y^2 + 1)^2$.
- **Take the Square Root:**
- • $\sqrt{(1 + [dx/dy]^2)} = y^2 + 1$.
- **Evaluate the Definite Integral from y = 0 to y = 1:**
- • $L = \int_0^1 (y^2 + 1)\,dy = [(1/3)y^3 + y]_0^1 = (1/3 + 1) - 0 = 4/3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Assumes endpoint change $\Delta y = 1$ gives arc length.

✗ **(B)** Forgets linear term y in antiderivative: 1/3 or miscalculates to 2/3.

✗ **(C)** Arithmetic error in adding 1/3 + 1: 5/3.

Question 85 Analysis

★★★★ MANDATORY

Correct Answer: (A) 6π

STEP-BY-STEP SOLUTION

- **Parametric Arc Length / Distance Traveled Formula:** Distance = $\int_0^{2\pi} \sqrt{([x'(t)]^2 + [y'(t)]^2)}\, dt$.
- **Differentiate Components:**
- • $x'(t) = -3 \sin t$.
- • $y'(t) = 3 \cos t$.
- **Compute Speed:**
- • Speed = $\sqrt{((-3 \sin t)^2 + (3 \cos t)^2)} = \sqrt{(9 \sin^2 t + 9 \cos^2 t)} = \sqrt{9} = 3$.
- **Integrate Speed Over Time Interval [0, 2π]:**
- • Distance = $\int_0^{2\pi} 3\, dt = 3 \cdot (2\pi - 0) = 6\pi$.
- **Geometric Check:** Circle of radius r = 3 has circumference $C = 2\pi r = 6\pi$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Integrates over a half period [0, π]: 3π.

✗ **(C)** Calculates the area of the circle $\pi r^2 = 9\pi$ instead of distance traveled.

✗ **(D)** Doubles the circumference: 12π.

Question 86 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) (8 / 27) (10√10 – 1)

STEP-BY-STEP SOLUTION

- **Distance Traveled Formula:** Distance = $\int_0^2 \sqrt{(}(x'(t))^2 + (y'(t))^2\,)\, dt$.
- **Compute Speed:**
- • $x'(t) = 2t$, $y'(t) = 3t^2$.
- • Speed = $\sqrt{(4t^2 + 9t^4)} = \sqrt{(}\, t^2(4 + 9t^2)\,) = t\sqrt{(4 + 9t^2)}$ (since $t \geq 0$).
- **Evaluate the Definite Integral using Substitution:**
- • Let $u = 4 + 9t^2 \Rightarrow du = 18t\, dt \Rightarrow t\, dt = du / 18$.
- • At t = 0: $u = 4 + 0 = 4$.
- • At t = 2: $u = 4 + 9(4) = 4 + 36 = 40$.
- • Distance = (1 / 18) $\int_4^{40} u\text{^}(1/2)\, du$ = (1 / 18) $[(2/3)\, u\text{^}(3/2)\,]_4^{40}$.
- • (1 / 27) [40^(3/2) – 4^(3/2)].
- • $40\text{^}(3/2) = 40\sqrt{40} = 40(2\sqrt{10}) = 80\sqrt{10}$.
- • $4\text{^}(3/2) = 8$.
- • (1 / 27) [80√10 – 8] = (8 / 27) (10√10 – 1).

WHY OTHER OPTIONS ARE INCORRECT

Question 87 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) ln(2 + √3)

STEP-BY-STEP SOLUTION

- **Differentiate y = ln(cos x):**
- • $y' = (1 / \cos x) \cdot (-\sin x) = -\tan x$.
- **Compute $1 + (y')^2$:**
- • $1 + (-\tan x)^2 = 1 + \tan^2 x = \sec^2 x$.
- • $\sqrt{(1 + (y')^2)} = \sqrt{(\sec^2 x)} = \sec x$ (since sec x > 0 on [0, π/3]).
- **Evaluate the Definite Integral:**
- • $L = \int_0^{\pi/3} \sec x\, dx$ = [ln |sec x + tan x|]₀^(π/3).
- • At $x = \pi/3$: $\sec(\pi/3) = 2$, $\tan(\pi/3) = \sqrt{3} \Rightarrow \ln(2 + \sqrt{3})$.
- • At x = 0: $\sec(0) = 1$, $\tan(0) = 0 \Rightarrow \ln(1 + 0) = \ln(1) = 0$.
- • $L = \ln(2 + \sqrt{3}) - 0 = \ln(2 + \sqrt{3})$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Neglects the tan x term in antiderivative of sec x: ln 2.

✗ **(B)** Uses $\sec(\pi/3) = 1$ instead of 2: $\ln(1 + \sqrt{3})$.

✗ **(D)** Doubles the correct value: $2 \ln(2 + \sqrt{3})$.

Question 88 Analysis

★★★★ MANDATORY

Correct Answer: (D) 3

STEP-BY-STEP SOLUTION

- **Compute the Particle's Speed:**
- • Speed = $\sqrt{([x'(t)]^2 + [y'(t)]^2)} = \sqrt{((e^t \cos t)^2 + (e^t \sin t)^2)}$.
- • $\sqrt{(e^{2t}\cos^2 t + e^{2t}\sin^2 t)} = \sqrt{(e^{2t}(\cos^2 t + \sin^2 t))} = \sqrt{(e^{2t})} = e^t$.
- **Set Up and Evaluate Distance Integral:**
- • Distance = $\int_0^{\ln 4} e^t\,dt = [e^t]_0^{(\ln 4)}$.
- • At $t = \ln 4$: $e^{\ln 4} = 4$.
- • At $t = 0$: $e^0 = 1$.
- • Distance = $4 - 1 = 3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Square roots the result: $\sqrt{4} = 2$.
✗ **(B)** Forgets to subtract lower limit evaluation $e^0 = 1$: 4.
✗ **(C)** Confuses total distance with the time elapsed $\Delta t = \ln 4$.

Question 89 Analysis

★★★★ MANDATORY

Correct Answer: (A) 12

STEP-BY-STEP SOLUTION

- **Arc Length Formula:** $L = \int_0^4 \sqrt{(1 + [f'(x)]^2)}\,dx$.
- **Compute $g(x) = \sqrt{(1 + [f'(x)]^2)}$ at Each Given x:**
- • At $x = 0$: $\sqrt{(1 + 0^2)} = \sqrt{1} = 1$.
- • At $x = 1$: $\sqrt{(1 + (\sqrt{3})^2)} = \sqrt{(1 + 3)} = \sqrt{4} = 2$.
- • At $x = 2$: $\sqrt{(1 + (\sqrt{8})^2)} = \sqrt{(1 + 8)} = \sqrt{9} = 3$.
- • At $x = 3$: $\sqrt{(1 + (\sqrt{15})^2)} = \sqrt{(1 + 15)} = \sqrt{16} = 4$.
- • At $x = 4$: $\sqrt{(1 + (\sqrt{24})^2)} = \sqrt{(1 + 24)} = \sqrt{25} = 5$.
- **Apply Trapezoidal Rule with $\Delta x = 1$:**
- • $L \approx (\Delta x / 2) \cdot [\,g(0) + 2g(1) + 2g(2) + 2g(3) + g(4)\,]$.
- • $(1/2) \cdot [\,1 + 2(2) + 2(3) + 2(4) + 5\,]$.
- • $(1/2) \cdot [\,1 + 4 + 6 + 8 + 5\,] = (1/2) \cdot 24 = 12$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets to divide by 2 in the trapezoidal rule formula: 24.
✗ **(C)** Uses a Left Riemann sum: $1 \cdot (1 + 2 + 3 + 4) = 10$.

Question 90 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) Area < V_1 < L

STEP-BY-STEP SOLUTION

- **1. Compute Area of R:**
- • Area = $\int_0^1 x^2\,dx = [x^3/3]_0^1 = 1/3 \approx 0.333$.
- **2. Compute Volume V_1 of Revolution about x-axis:**
- • $V_1 = \pi \int_0^1 (x^2)^2\,dx = \pi \int_0^1 x^4\,dx = \pi\,[x^5/5]_0^1 = \pi/5 \approx 3.14159/5 \approx 0.628$.
- **3. Estimate/Compute Arc Length L of $y = x^2$ on [0, 1]:**
- • $L = \int_0^1 \sqrt{(1 + (2x)^2)}\,dx = \int_0^1 \sqrt{(1 + 4x^2)}\,dx$.
- • The straight-line distance between (0, 0) and (1, 1) is $\sqrt{(1^2 + 1^2)} = \sqrt{2} \approx 1.414$.
- • Since the curve is curved, $L > \sqrt{2} \approx 1.414$ (exact value is ≈ 1.4789).
- **Compare Values:**
- • Area $\approx 0.333 < V_1 \approx 0.628 < L \approx 1.479$.
- • Thus, Area < V_1 < L.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Believes V_1 < Area because $\pi/5$ is compared incorrectly to 1/3.
✗ **(C)** Incorrectly assumes volume exceeds arc length in magnitude.
✗ **(D)** Believes arc length is shorter than area.

FRQ 1: Particle Motion on a Coordinate Line (Displacement, Distance, Tables & Turning Points) Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (A): Acceleration Estimate at t = 6:

- Acceleration is the derivative of velocity: $a(6) = v'(6)$.
- Using the interval [5, 7] containing t = 6:
- $a(6) \approx (v(7) - v(5)) / (7 - 5) = (-4 - 2) / (7 - 5) = -6 / 2 = -3$ m/s^2.
- Units: meters per second squared (m/s^2).

Part (B): Trapezoidal Sum Approximation for Displacement:

- Displacement = $\int_0^{10} v(t)\,dt$.
- Subinterval [0, 2]: $\Delta t = 2$. $\text{Area}_1 = (2/2) \cdot (5 + 8) = 1 \cdot 13 = 13$.
- Subinterval [2, 5]: $\Delta t = 3$. $\text{Area}_2 = (3/2) \cdot (8 + 2) = 1.5 \cdot 10 = 15$.
- Subinterval [5, 7]: $\Delta t = 2$. $\text{Area}_3 = (2/2) \cdot (2 + (-4)) = 1 \cdot (-2) = -2$.
- Subinterval [7, 10]: $\Delta t = 3$. $\text{Area}_4 = (3/2) \cdot (-4 + (-1)) = 1.5 \cdot (-5) = -7.5$.
- Total Displacement $\approx 13 + 15 - 2 - 7.5 = 18.5$ meters.

Part (C): Exact Position at t = 2:

- $x(2) = x(0) + \int_0^2 v_1(t)\,dt$.
- Given $x(0) = -3$ and $v_1(t) = 5 + 3t - (3/4)t^2$.
- $\int_0^2 (5 + 3t - (3/4)t^2)\,dt = [\,5t + (3/2)t^2 - (1/4)t^3\,]_0^2$.
- At t = 2: $5(2) + (3/2)(4) - (1/4)(8) = 10 + 6 - 2 = 14$.
- Thus, $x(2) = -3 + 14 = 11$ meters.

Part (D): Justification of Rest and Total Distance:

- Since v(t) is differentiable, it is continuous on [2, 7].
- From the table, $v(2) = 8 > 0$ and $v(7) = -4 < 0$.
- By the Intermediate Value Theorem (IVT), since 0 lies between −4 and 8, there must exist at least one time $t \in (2, 7)$ where $v(t) = 0$ (particle is at rest).
- Because v(t) is positive on [0, 2] and negative on [7, 10], the particle changes direction.
- When velocity changes sign, $\int_0^{10} |v(t)|\,dt > \left|\int_0^{10} v(t)\,dt\right|$.
- Therefore, the total distance traveled is strictly greater than the displacement.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (A)	1 pt for difference quotient setup; 1 pt for estimate with correct units (m/s^2).	**2 pts**
Part (B)	1 pt for trapezoidal sum terms; 1 pt for approximation of displacement.	**2 pts**
Part (C)	1 pt for definite integral setup; 1 pt for antiderivative; 1 pt for final position x(2) = 11.	**3 pts**
Part (D)	1 pt for IVT justification; 1 pt for reasoning regarding total distance > displacement.	**2 pts**

FRQ 2: Water Storage Inflow and Outflow Rates (Net Change & Optimization) Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (A): Total Inflow from t = 0 to t = 8:

- Total Inflow = $\int_0^8 R(t)\,dt = \int_0^8 (60 - 6t)\,dt$.
- Antiderivative: $[\,60t - 3t^2\,]_0^8$.
- At t = 8: $60(8) - 3(8)^2 = 480 - 3(64) = 480 - 192 = 288$ gallons.

Part (B): Behavior at t = 5:

- Net rate of change of water is $W'(t) = R(t) - D(t)$.
- At t = 5: $R(5) = 60 - 6(5) = 30$ gal/min.
- Outflow is constant: $D(5) = 24$ gal/min.
- Net rate: $W'(5) = R(5) - D(5) = 30 - 24 = +6$ gal/min > 0.
- Since $W'(5) > 0$, the amount of water in the reservoir is increasing at t = 5.

Part (C): Absolute Maximum Amount of Water on [0, 8]:

- Volume function: $W(t) = 200 + \int_0^t [R(u) - D(u)]\,du = 200 + \int_0^t (36 - 6u)\,du$.
- Critical Points: Set $W'(t) = 0 \Rightarrow 36 - 6t = 0 \Rightarrow t = 6$ minutes.
- Evaluate W(t) at critical point and endpoints (Candidates Test):
- 1. At t = 0: $W(0) = 200$ gallons.
- 2. At t = 6: $W(6) = 200 + [\,36t - 3t^2\,]_0^6 = 200 + (36(6) - 3(36)) = 200 + (216 - 108) = 200 + 108 = 308$ gallons.
- 3. At t = 8: $W(8) = 200 + [\,36(8) - 3(64)\,] = 200 + (288 - 192) = 200 + 96 = 296$ gallons.
- Conclusion: The absolute maximum amount of water in the reservoir is 308 gallons, occurring at t = 6 minutes.

Part (D): Average Rate of Inflow Expression:

- $R_{avg} = (1/(8-0)) \int_0^8 R(t)\,dt = (1/8) \int_0^8 (60 - 6t)\,dt$.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (A)	1 pt for definite integral setup $\int_0^8 R(t)\,dt$; 1 pt for evaluation (288 gallons).	**2 pts**
Part (B)	1 pt for comparing R(5) and D(5); 1 pt for conclusion with reason (increasing).	**2 pts**
Part (C)	1 pt for identifying critical point t = 6; 1 pt for candidates test; 1 pt for maximum amount (308 gallons).	**3 pts**
Part (D)	1 pt for average value formula (1/8); 1 pt for correct integral expression.	**2 pts**

FRQ 3: Planar Region Bounded by Curves, Area & Dividing Lines Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (A): Points of Intersection and Area of R:

- Set $f(x) = g(x) \Rightarrow 2\sqrt{x} = (1/2)x^2 \Rightarrow 4\sqrt{x} = x^2$.
- Square both sides: $16x = x^4 \Rightarrow x(x^3 - 16) = 0$.
- Wait: let's re-verify $2\sqrt{x} = x^2/2$: $4\sqrt{x} = x^2 \Rightarrow 16x = x^4$. Wait, $4\sqrt{4} = 4 \cdot 2 = 8$, $4^2/2 = 16/2 = 8$. No, when $x = 4$: $2\sqrt{4} = 4$ and $4^2/2 = 8$ (not equal!).
- Wait, if $g(x) = x^2/4$: $2\sqrt{4} = 4$ and $4^2/4 = 4$! That intersects at (4, 4).
- For $f(x) = 2\sqrt{x}$ and $g(x) = (1/2)x^2$: $2\sqrt{x} = x^2/2 \Rightarrow 4\sqrt{x} = x^2 \Rightarrow 16x = x^4 \Rightarrow x = 0$ or $x^3 = 16 \Rightarrow x = 16\text{^}(1/3)$.
- Let's adjust $g(x)$ to $g(x) = x^2/4$ in the stem! Then $2\sqrt{x} = x^2/4 \Rightarrow 8\sqrt{x} = x^2 \Rightarrow 64x = x^4 \Rightarrow x(x^3 - 64) = 0 \Rightarrow x = 0$ and $x = 4$. That is clean and elegant!
- On [0, 4], $f(x) \geq g(x)$: $f(x) = 2\sqrt{x}$, $g(x) = x^2/4$.
- Area = $\int_0^4 (2x\text{^}(1/2) - x^2/4)\,dx = [(4/3)x\text{^}(3/2) - x^3/12]_0^4$.
- At $x = 4$: $(4/3)(8) - 64/12 = 32/3 - 16/3 = 16/3$.

Part (B): Vertical Dividing Line x = k:

- Total area = 16/3, so half of the area is $(1/2) \cdot (16/3) = 8/3$.
- The equation for k is:
- $\int_0^k (2\sqrt{x} - x^2/4)\,dx = 8/3$ (or $\int_0^k (2\sqrt{x} - x^2/4)\,dx = \int_k^4 (2\sqrt{x} - x^2/4)\,dx$).

Part (C): Area Integrating with Respect to y:

- Express curves as x in terms of y:
- $f(x) = 2\sqrt{x} \Rightarrow \sqrt{x} = y/2 \Rightarrow x_left = y^2/4$.
- $g(x) = x^2/4 \Rightarrow x^2 = 4y \Rightarrow x_right = 2\sqrt{y}$.
- As x ranges from 0 to 4, y ranges from 0 to 4.
- Horizontal slicing: Area = $\int_0^4 (2\sqrt{y} - y^2/4)\,dy$.
- $[(4/3)y\text{^}(3/2) - y^3/12]_0^4 = (4/3)(8) - 64/12 = 32/3 - 16/3 = 16/3$.
- The two calculations yield the identical area of 16/3.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (A)	1 pt for intersection points (0, 0) and (4, 4); 1 pt for antiderivative; 1 pt for area = 16/3.	**3 pts**
Part (B)	1 pt for limits [0, k]; 1 pt for integrand $f(x) - g(x)$; 1 pt for equating to 8/3 (or half of total area).	**3 pts**
Part (C)	1 pt for expressing curves as functions of y; 1 pt for limits in y [0, 4]; 1 pt for evaluation to 16/3.	**3 pts**

FRQ 4: Volumes of Solids with Known Cross Sections Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (A): Square Cross Sections:

- Side length of each square is $s(x) = (4 - x^2) - 0 = 4 - x^2$.
- Area of square cross section: $A(x) = [s(x)]^2 = (4 - x^2)^2 = 16 - 8x^2 + x^4$.
- Volume = $\int_{-2}^{2} (16 - 8x^2 + x^4)\,dx = 2\int_{0}^{2} (16 - 8x^2 + x^4)\,dx$.
- $2\,[16x - (8/3)x^3 + (1/5)x^5\,]_0^2$.
- At $x = 2$: $16(2) - (8/3)(8) + (1/5)(32) = 32 - 64/3 + 32/5$.
- $32 \cdot [\,1 - 2/3 + 1/5\,] = 32 \cdot [\,8/15\,] = 256/15$.
- Volume $= 2 \cdot (256/15) = 512/15$.

Part (B): Equilateral Triangle Cross Sections:

- Area of equilateral triangle with side s: $A(x) = (\sqrt{3}/4) \cdot s^2$.
- Here, $s(x) = 4 - x^2$, so $A(x) = (\sqrt{3}/4) \cdot (4 - x^2)^2$.
- Volume $= (\sqrt{3}/4)\int_{-2}^{2} (4 - x^2)^2\,dx$.
- From Part (A), the integral value is $512/15$.
- Volume $= (\sqrt{3}/4) \cdot (512/15) = 128\sqrt{3}/15$.

Part (C): Semicircle Cross Sections:

- Diameter $d(x) = 4 - x^2 \Rightarrow$ Radius $r(x) = (4 - x^2)/2$.
- Area of semicircle: $A(x) = (1/2)\pi\,[r(x)]^2 = (1/2)\pi\,[\,(4 - x^2)/2\,]^2 = (\pi/8)(4 - x^2)^2$.
- Volume $= (\pi/8)\int_{-2}^{2} (4 - x^2)^2\,dx$.
- Volume $= (\pi/8) \cdot (512/15) = 64\pi/15$.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (A)	1 pt for integrand $(4 - x^2)^2$; 1 pt for symmetry/antiderivative; 1 pt for volume = 512/15.	**3 pts**
Part (B)	1 pt for constant factor $(\sqrt{3}/4)$; 1 pt for integral setup; 1 pt for volume = $128\sqrt{3}/15$.	**3 pts**
Part (C)	1 pt for radius $r = (4 - x^2)/2$; 1 pt for constant factor $(\pi/8)$; 1 pt for volume = $64\pi/15$.	**3 pts**

FRQ 5: Volumes of Revolution: Washer Method around Horizontal & Vertical Axes Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (A): Revolved About the x-axis (y = 0):

- Intersection points: $2x = x^2 \Rightarrow x = 0$ and $x = 2$.
- For $0 < x < 2$, outer radius $R(x) = 2x$ and inner radius $r(x) = x^2$.
- Volume $= \pi \int_0^2 [(2x)^2 - (x^2)^2]\,dx = \pi \int_0^2 (4x^2 - x^4)\,dx$.
- $\pi [(4/3)x^3 - (1/5)x^5]_0^2$.
- At $x = 2$: $(4/3)(8) - (1/5)(32) = 32/3 - 32/5 = 32 \cdot (1/3 - 1/5) = 32 \cdot (2/15) = 64/15$.
- Volume $= 64\pi / 15$.

Part (B): Revolved About the Horizontal Line y = 5:

- The line $y = 5$ lies above both curves on $[0, 2]$.
- Outer radius (distance to farther curve $y = x^2$): $R(x) = 5 - x^2$.
- Inner radius (distance to closer curve $y = 2x$): $r(x) = 5 - 2x$.
- $[R(x)]^2 - [r(x)]^2 = (5 - x^2)^2 - (5 - 2x)^2 = (25 - 10x^2 + x^4) - (25 - 20x + 4x^2) = x^4 - 14x^2 + 20x$.
- Volume $= \pi \int_0^2 (x^4 - 14x^2 + 20x)\,dx = \pi [(1/5)x^5 - (14/3)x^3 + 10x^2]_0^2$.
- At $x = 2$: $32/5 - 112/3 + 40 = 32/5 + 8/3 = 136/15$.
- Volume $= 136\pi / 15$.

Part (C): Revolved About the Vertical Line x = 3:

- Revolving about a vertical line requires horizontal slicing (integrating with respect to y).
- Limits in y: y ranges from 0 to 4.
- Express curves as x in terms of y:
- Line: $y = 2x \Rightarrow x = y / 2$ (closer to line $x = 3$ for lower curve).
- Parabola: $y = x^2 \Rightarrow x = \sqrt{y}$ (farther from line $x = 3$ for given y).
- Wait, on $[0, 4]$, $x = y/2 \le \sqrt{y}$. Since axis is $x = 3$:
- Outer radius $R(y) = 3 - (y / 2)$ (farther from $x = 3$).
- Inner radius $r(y) = 3 - \sqrt{y}$ (closer to $x = 3$).
- Volume $= \pi \int_0^4 [(3 - y/2)^2 - (3 - \sqrt{y})^2]\,dy$.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (A)	1 pt for washer integrand $(2x)^2 - (x^2)^2$; 1 pt for antiderivative; 1 pt for volume $= 64\pi / 15$.	**3 pts**
Part (B)	1 pt for outer/inner radii $R = 5 - x^2$, $r = 5 - 2x$; 1 pt for difference of squares; 1 pt for volume $= 136\pi / 15$.	**3 pts**
Part (C)	1 pt for integrating in y with limits $[0, 4]$; 1 pt for outer radius $(3 - y/2)$; 1 pt for inner radius $(3 - \sqrt{y})$.	**3 pts**

FRQ 6: BC Exclusive: Arc Length, Perimeter & Volume of a Planar Solid Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (A): Area of Region R:

- Area = $\int_0^3 (2/3)\,x^{(3/2)}\,dx = [\,(2/3)\cdot(2/5)\,x^{(5/2)}\,]_0^3 = [\,(4/15)\,x^{(5/2)}\,]_0^3$.
- At $x = 3$: $(4/15)\cdot 3^{(5/2)} = (4/15)\cdot 9\sqrt{3} = (36\sqrt{3})/15 = (12\sqrt{3})/5$.

Part (B): Perimeter of Region R:

- Region R has three boundaries:
- 1. Bottom straight boundary along x-axis from $x = 0$ to $x = 3$: Length = 3.
- 2. Right vertical boundary at $x = 3$ from $y = 0$ to $y = (2/3)(3)^{(3/2)} = (2/3)(3\sqrt{3}) = 2\sqrt{3}$: Length = $2\sqrt{3}$.
- 3. Curved boundary along $y = (2/3)\,x^{(3/2)}$ from $x = 0$ to $x = 3$:
- $f'(x) = \sqrt{x} \Longrightarrow 1 + [f'(x)]^2 = 1 + x$.
- Arc Length $L = \int_0^3 \sqrt{(1+x)}\,dx = [\,(2/3)(1+x)^{(3/2)}\,]_0^3$.
- $L = (2/3)[\,(4)^{(3/2)} - (1)^{(3/2)}\,] = (2/3)[\,8 - 1\,] = 14/3$.
- Total Perimeter $= 3 + 2\sqrt{3} + 14/3 = (9/3 + 14/3) + 2\sqrt{3} = 23/3 + 2\sqrt{3}$.

Part (C): Volume of Revolution About x-axis:

- Disk method with radius $R(x) = (2/3)\,x^{(3/2)}$:
- Volume $= \pi \int_0^3 [\,(2/3)\,x^{(3/2)}\,]^2\,dx = \pi \int_0^3 (4/9)\,x^3\,dx$.
- $\pi\cdot(4/9)\,[\,x^4/4\,]_0^3 = (\pi/9)\,[\,x^4\,]_0^3$.
- At $x = 3$: $(\pi/9)(81) = 9\pi$.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (A)	1 pt for antiderivative; 1 pt for area = $(4\sqrt{3})/5$.	**2 pts**
Part (B)	1 pt for derivative $f'(x) = \sqrt{x}$; 1 pt for arc length integral $\int_0^3 \sqrt{(1+x)}\,dx$; 1 pt for arc length = 14/3; 1 pt for total perimeter = $23/3 + 2\sqrt{3}$.	**4 pts**
Part (C)	1 pt for disk method setup $\pi\int (4/9)x^3\,dx$; 1 pt for antiderivative; 1 pt for volume = 9π.	**3 pts**

FRQ 7: Comprehensive Solid Modeling - Cross Sections vs. Revolution Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (A): Area of Region R:

- Intersection: $e^x = 1 \Rightarrow x = 0$. The region extends from $x = 0$ to $x = \ln 4$.
- Area = $\int_0^{\ln 4} (e^x - 1)\,dx = [\,e^x - x\,]_0^{(\ln 4)}$.
- At $x = \ln 4$: $e^{\ln 4} - \ln 4 = 4 - \ln 4$.
- At $x = 0$: $e^0 - 0 = 1$.
- Area = $(4 - \ln 4) - 1 = 3 - \ln 4$.

Part (B): Solid with Rectangular Cross Sections:

- Base of rectangle in xy-plane: $b(x) = e^x - 1$.
- Height of rectangle: $h(x) = 2 \cdot b(x) = 2(e^x - 1)$.
- Cross-sectional area: $A(x) = b(x) \cdot h(x) = 2(e^x - 1)^2 = 2(e^{2x} - 2e^x + 1)$.
- Volume = $\int_0^{\ln 4} 2(e^{2x} - 2e^x + 1)\,dx = 2\,[\,(1/2)\,e^{2x} - 2e^x + x\,]_0^{(\ln 4)}$.
- At $x = \ln 4$: $(1/2)(16) - 2(4) + \ln 4 = 8 - 8 + \ln 4 = \ln 4$.
- At $x = 0$: $(1/2)(1) - 2(1) + 0 = 1/2 - 2 = -3/2$.
- Value inside brackets: $\ln 4 - (-3/2) = 3/2 + \ln 4$.
- Multiply by 2: Volume = $2(3/2 + \ln 4) = 3 + 2\ln 4$.

Part (C): Volume of Revolution About y = 1:

- The axis of revolution is the lower boundary $y = 1$.
- Circular disk radius: $R(x) = e^x - 1$.
- Volume = $\pi \int_0^{\ln 4} (e^x - 1)^2\,dx = \pi \int_0^{\ln 4} (e^{2x} - 2e^x + 1)\,dx$.
- Notice that this integral is exactly half of the integral computed in Part (B) multiplied by π!
- Volume = $\pi \cdot (3/2 + \ln 4) = \pi(3 + 2\ln 4)\,/\,2$.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (A)	1 pt for integral setup $\int_0^{\ln 4} (e^x - 1)\,dx$; 1 pt for area = $3 - \ln 4$.	**2 pts**
Part (B)	1 pt for cross-sectional area $A(x) = 2(e^x - 1)^2$; 1 pt for antiderivative; 1 pt for volume = $15 - 4\ln 4$.	**3 pts**
Part (C)	1 pt for disk radius $R(x) = e^x - 1$; 1 pt for integral $\pi \int (e^x - 1)^2\,dx$; 1 pt for antiderivative; 1 pt for volume = $\pi(15/2 - 2\ln 4)$.	**4 pts**

FRQ 8: BC Exclusive: Planar Motion, Parametric Arc Length & Distance Traveled Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (A): Speed at t = 2:

- Speed = $\sqrt{([x'(t)]^2 + [y'(t)]^2)}$.
- Given $x'(t) = 4t$ and $y'(t) = 3t^2$.
- At $t = 2$: $x'(2) = 8$ and $y'(2) = 12$.
- Speed(2) = $\sqrt{(8^2 + 12^2)} = \sqrt{(64 + 144)} = \sqrt{208} = \sqrt{(16 \cdot 13)} = 4\sqrt{13}$.

Part (B): Acceleration Vector at t = 2:

- $a(t) = \langle x''(t), y''(t) \rangle = \langle d/dt(4t), d/dt(3t^2) \rangle = \langle 4, 6t \rangle$.
- At $t = 2$: $a(2) = \langle 4, 6(2) \rangle = \langle 4, 12 \rangle$.

Part (C): Total Distance Traveled on [0, 3]:

- Distance = $\int_0^3 \sqrt{((4t)^2 + (3t^2)^2)}\, dt = \int_0^3 \sqrt{(16t^2 + 9t^4)}\, dt$.
- Factor out t: Distance = $\int_0^3 t\sqrt{(16 + 9t^2)}\, dt$.
- Substitute $u = 16 + 9t^2 \Longrightarrow du = 18t\, dt \Longrightarrow t\, dt = du / 18$.
- At $t = 0$: $u = 16$.
- At $t = 3$: $u = 16 + 9(9) = 16 + 81 = 97$.
- Distance = (1/18) $\int_{16}^{97} u^{(1/2)}\, du$ = (1/18) [(2/3) $u^{(3/2)}$]$_{16}^{97}$ = (1/27) [$97^{(3/2)} - 16^{(3/2)}$].
- $16^{(3/2)} = 64$.
- Distance = (1 / 27) (97√97 − 64).

Part (D): Position at t = 3:

- $x(3) = x(0) + \int_0^3 x'(t)\, dt = 2 + \int_0^3 4t\, dt = 2 + [\,2t^2\,]_0^3 = 2 + 18 = 20$.
- $y(3) = y(0) + \int_0^3 y'(t)\, dt = -5 + \int_0^3 3t^2\, dt = -5 + [\,t^3\,]_0^3 = -5 + 27 = 22$.
- Position at $t = 3$ is (20, 22).

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (A)	1 pt for speed expression $\sqrt{(x'^2 + y'^2)}$; 1 pt for evaluation (speed = $4\sqrt{13}$).	**2 pts**
Part (B)	1 pt for component derivatives; 1 pt for vector $\langle 4, 12 \rangle$.	**2 pts**
Part (C)	1 pt for distance integral setup $\int_0^3 t\sqrt{(16 + 9t^2)}\, dt$; 1 pt for u-substitution; 1 pt for distance = 301/27.	**3 pts**
Part (D)	1 pt for x(3) = 20; 1 pt for y(3) = 22, yielding (20, 22).	**2 pts**

FRQ 9: Applied Accumulation & Geometric Rate Function Analysis Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (A): Total Fuel Consumed on [0, 10]:

- Total consumption = $\int_0^{10} r(t)\,dt$ = Area under the graph of r(t).
- The graph forms a trapezoid with height h = 12, bottom base $b_1 = 10$, and top base $b_2 = 7 - 3 = 4$.
- Area = $(1/2) \cdot (b_1 + b_2) \cdot h = (1/2) \cdot (10 + 4) \cdot 12 = 7 \cdot 12 = 84$ liters.

Part (B): Fuel Remaining at t = 7:

- Fuel remaining $F(7) = 150 - \int_0^7 r(t)\,dt$.
- Area from t = 0 to t = 7 decomposes into:
- Triangle from t = 0 to t = 3: $(1/2) \cdot 3 \cdot 12 = 18$ liters.
- Rectangle from t = 3 to t = 7: $(7 - 3) \cdot 12 = 4 \cdot 12 = 48$ liters.
- Total fuel consumed by t = 7: 18 + 48 = 66 liters.
- Fuel remaining = 150 – 66 = 84 liters.

Part (C): Average Rate of Fuel Consumption on [0, 10]:

- $r_{avg} = (1/10) \int_0^{10} r(t)\,dt = (1/10) \cdot 84 = 8.4$ liters per hour.

Part (D): Time where r(t) = 8.4 L/hr:

- On the interval [0, 3], the line connects (0, 0) and (3, 12).
- Equation of line: $r(t) = (12/3)\,t = 4t$.
- Set $4t = 8.4 \Rightarrow t = 8.4/4 = 2.1$ hours.
- (Alternatively, on [7, 10], the line connects (7, 12) and (10, 0): slope = –4, $r(t) = 12 - 4(t - 7) = 40 - 4t$. $40 - 4t = 8.4 \Rightarrow 4t = 31.6 \Rightarrow t = 7.9$ hours.)
- Thus, r(t) equals the average rate at t = 2.1 hours and t = 7.9 hours.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (A)	1 pt for trapezoidal area decomposition; 1 pt for total consumption = 84 liters.	**2 pts**
Part (B)	1 pt for consumption from 0 to 7 (66 L); 1 pt for fuel remaining = 84 liters.	**2 pts**
Part (C)	1 pt for average value formula $(1/10)\int r(t)\,dt$; 1 pt for calculation; 1 pt for 8.4 L/hr with units.	**3 pts**
Part (D)	1 pt for linear equation on [0, 3] or [7, 10]; 1 pt for finding t = 2.1 hr (or t = 7.9 hr).	**2 pts**

FRQ 10: Full Unit 8 Capstone (Area, Washer Volume, Cross Sections & Arc Length) Official Model Solution

★★★★★ VERY IMPORTANT

STEP-BY-STEP ANALYTICAL SOLUTION

Part (A): Points of Intersection and Area of R:

- Set $f(x) = g(x) \Rightarrow 6 - x^2 = x + 4 \Rightarrow x^2 + x - 2 = 0$.
- $(x + 2)(x - 1) = 0 \Rightarrow x = -2$ and $x = 1$.
- Intersection points are $(-2, 2)$ and $(1, 5)$.
- On $[-2, 1]$, $6 - x^2 \geq x + 4$.
- Area = $\int_{-2}^{1} [(6 - x^2) - (x + 4)]\, dx = \int_{-2}^{1} (2 - x - x^2)\, dx$.
- $[2x - (1/2)x^2 - (1/3)x^3]_{-2}^{1}$.
- At $x = 1$: $2(1) - 1/2 - 1/3 = 2 - 5/6 = 7/6$.
- At $x = -2$: $2(-2) - (1/2)(4) - (1/3)(-8) = -4 - 2 + 8/3 = -6 + 8/3 = -10/3$.
- Area = $7/6 - (-10/3) = 7/6 + 20/6 = 27/6 = 9/2$.

Part (B): Volume of Revolution About y = 7:

- Axis $y = 7$ lies above both curves for all $x \in [-2, 1]$.
- Outer radius (distance to farther line $g(x) = x + 4$):
- $R(x) = 7 - (x + 4) = 3 - x$.
- Inner radius (distance to closer parabola $f(x) = 6 - x^2$):
- $r(x) = 7 - (6 - x^2) = 1 + x^2$.
- $[R(x)]^2 - [r(x)]^2 = (3 - x)^2 - (1 + x^2)^2 = (9 - 6x + x^2) - (1 + 2x^2 + x^4) = 8 - 6x - x^2 - x^4$.
- Volume = $\pi \int_{-2}^{1} (8 - 6x - x^2 - x^4)\, dx$.
- $\pi [8x - 3x^2 - (1/3)x^3 - (1/5)x^5]_{-2}^{1}$.
- At $x = 1$: $8 - 3 - 1/3 - 1/5 = 5 - 8/15 = 67/15$.
- At $x = -2$: $8(-2) - 3(4) - (1/3)(-8) - (1/5)(-32) = -16 - 12 + 8/3 + 32/5 = -28 + 40/15 + 96/15 = -28 + 136/15 = -420/15 + 136/15 = -284/15$.
- Difference: $67/15 - (-284/15) = 351/15 = 117/5$.
- Volume = $117\pi/5$.

Part (C): Square Cross Sections Integral:

- Side length of square: $s(x) = f(x) - g(x) = 2 - x - x^2$.
- Area of square: $A(x) = [s(x)]^2 = (2 - x - x^2)^2$.
- Volume = $\int_{-2}^{1} (2 - x - x^2)^2\, dx$.

Part (D): Perimeter of Region R:

- Boundary 1 (Straight line segment along $y = x + 4$ from $(-2, 2)$ to $(1, 5)$):
- Length = $\sqrt{((1 - (-2))^2 + (5 - 2)^2)} = \sqrt{(3^2 + 3^2)} = \sqrt{18} = 3\sqrt{2}$.
- Boundary 2 (Curved arc along $y = 6 - x^2$ from $x = -2$ to $x = 1$):
- $f'(x) = -2x \Rightarrow 1 + [f'(x)]^2 = 1 + (-2x)^2 = 1 + 4x^2$.
- Arc Length = $\int_{-2}^{1} \sqrt{(1 + 4x^2)}\, dx$.
- Perimeter = $3\sqrt{2} + \int_{-2}^{1} \sqrt{(1 + 4x^2)}\, dx$.

AP SCORING CRITERIA & POINT ALLOCATION

Part & Criterion	Scoring Guidelines	Points
Part (A)	1 pt for intersection $x = -2$ and $x = 1$; 1 pt for area = 9/2.	**2 pts**
Part (B)	1 pt for washer radii $R = 7 - (x + 4)$, $r = 7 - (6 - x^2)$; 1 pt for integrand; 1 pt for volume = $108\pi/5$.	**3 pts**
Part (C)	1 pt for side length $(2 - x - x^2)$; 1 pt for integral $\int_{-2}^{1} (2 - x - x^2)^2\, dx$.	**2 pts**
Part (D)	1 pt for arc length of parabola $\int_{-2}^{1} \sqrt{(1 + 4x^2)}\, dx$; 1 pt for straight line length $3\sqrt{2}$.	**2 pts**

Essential Unit 8 Rules

Core formulas, geometric area and volume theorems, and numerical methods required for AP Calculus BC.

- **Average Value of a Function:** $f_{avg} = (1 / (b - a)) \int_a^b f(x)\,dx$. By the Mean Value Theorem for Integrals, if f is continuous on [a, b], there exists $c \in (a, b)$ such that $f(c) = f_{avg}$.
- **Net Change Theorem & Particle Motion:** • Displacement: $\Delta x = \int_{t_1}^{t_2} v(t)\,dt$. Final Position: $x(t_2) = x(t_1) + \int_{t_1}^{t_2} v(t)\,dt$. • Total Distance Traveled: Total Distance = $\int_{t_1}^{t_2} |v(t)|\,dt$. • Speed = $|v(t)|$. Speed is increasing when v(t) and a(t) have the same sign ($v(t) \cdot a(t) > 0$); decreasing when they have opposite signs.
- **Area Between Curves:** • Slicing perpendicular to x-axis: Area = $\int_a^b [y_{top}(x) - y_{bottom}(x)]\,dx$. • Slicing perpendicular to y-axis: Area = $\int_c^d [x_{right}(y) - x_{left}(y)]\,dy$.
- **Volumes with Known Cross Sections:** Volume = $\int A(x)\,dx$ (or $\int A(y)\,dy$), where: • Squares: $A(x) = s^2$. • Equilateral Triangles: $A(x) = (\sqrt{3} / 4)\,s^2$. • Semicircles (diameter in base): $A(x) = (1/2)\,\pi\,(s / 2)^2 = (\pi / 8)\,s^2$. • Isosceles Right Triangles (hypotenuse in base): $A(x) = (1/4)\,s^2$. • Isosceles Right Triangles (leg in base): $A(x) = (1/2)\,s^2$.
- **Volumes of Revolution:** • **Disc Method:** Volume = $\pi \int [R(x)]^2\,dx$ (or $\pi \int [R(y)]^2\,dy$) when region directly abuts axis. • **Washer Method:** Volume = $\pi \int [(R(x))^2 - (r(x))^2]\,dx$, where R(x) is distance to outer curve, and r(x) is distance to inner curve. • **Shifted Horizontal Axis y = k:** $R(x) = |y_{outer} - k|$, $r(x) = |y_{inner} - k|$. • **Shifted Vertical Axis x = k:** $R(y) = |x_{outer} - k|$, $r(y) = |x_{inner} - k|$.
- **BC Exclusive: Arc Length of a Curve:** • As a function of x: $L = \int_a^b \sqrt{(1 + [f'(x)]^2)}\,dx$. • As a function of y: $L = \int_c^d \sqrt{(1 + [g'(y)]^2)}\,dy$.
- **BC Exclusive: Parametric Arc Length & Planar Distance Traveled:** • Distance Traveled = $\int_{t_1}^{t_2} \sqrt{((x'(t))^2 + (y'(t))^2)}\,dt = \int_{t_1}^{t_2} \text{Speed}(t)\,dt$.

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Thank You, Students!

Mastery Comes Through Deliberate Practice

Congratulations on completing this comprehensive 100-Question Priority-Starred Master Repository for Unit 8! Applications of integration form the capstone of single-variable calculus, bridging theoretical limits to geometric volumes, dynamic motion, and arc lengths. Review your steps, reflect on the distractor rationales, and approach your AP Calculus BC examination with absolute mastery and confidence!

TARGET SCORE: 5 · AP CALCULUS BC